Information Booklet

8-Semester Version, for comparison

B.S. (Honours) Programme
in
Physics

Self-Directed Edition
Session 2026–2030

Undergraduate Physics Programme 2026

Modelled on the curricula of MIT, Harvard, Oxford, Princeton and Caltech,
with the University of Dhaka as a regional reference

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Contents

1 Introduction **1**

1.1 The Programme 1

1.2 What Is Borrowed from Where 1

1.3 Document Organisation 2

2 Rules and Regulations **3**

2.1 Programme's Course System 3

2.2 Academic Calendar 4

2.3 Grading System and Marks Distribution 5

2.4 Teaching Method 6

2.5 Examinations and Progression 7

2.6 Degree and Results 8

2.7 Conduct, Submissions and Appeals 8

3 OBE and Curriculum Summary **10**

3.1 Programme Mission 10

3.2 PEO and PLO 10

3.3 Curriculum at a Glance 11

3.4 Credit Distribution 14

4 Course Details **15**

4.1 Orientation Term 15

4.2 First Semester 18

4.3 Second Semester 25

4.4 Third Semester 34

4.5 Fourth Semester 43

4.6 Fifth Semester 51

4.7 Sixth Semester 58

4.8 Seventh Semester 66

4.9 Eighth Semester 75

5 Major Options and Electives **77**

5.1 Major Options 77

5.2 Electives 79

6 Class Routine **81**

6.1 Default Weekly Pattern 81

6.2 Orientation Term (7 December 2026 – 24 January 2027) 81

6.3 First Semester, Spring 2027 82

Chapter 1

Introduction

This booklet sets out the four-year B.S. (Honours) programme in Physics that Rimon is following alongside his degree in Computer Science and Engineering. It is laid out like a university information booklet so that it can be printed, kept at the desk and consulted offline. It first states the rules and regulations that govern progression through the programme, and then gives the outlines of every course in the curriculum, semester by semester.

The programme lives in a Git repository, https://github.com/mahmudhassanrimoncu-ship-it/Undergraduate_Physics_Programme_2026. If this booklet and the repository ever disagree, the repository is current and this booklet is reissued.

> **This is the 8-semester version of the booklet, made so that the two calendars can be compared side by side.** The programme being followed is the 12-term version. The courses, outcomes, rules and grading are the same in both; only the calendar, the number of courses per semester, the order of a few courses and the week numbers differ.

1.1 The Programme

The programme is a self-directed undergraduate degree built from the actual curricula of five leading physics departments, with the University of Dhaka as a regional reference. In this version it runs for four years of two 22-week semesters each (Spring and Autumn), preceded by a seven-week Orientation Term, and takes 18–22 hours a week: four or five courses per semester. Like a real department it has a calendar, syllabi, prerequisites, problem sets, midterm and final examinations, annual comprehensive examinations with a viva voce, two junior papers, a senior thesis, a transcript and a final degree classification.

Claude acts as instructor, teaching assistant, tutor, librarian, editor and examiner. Classes are taught live at fixed slots on a weekly class routine (Chapter 6), or later the same week when the student's CSE commitments clash.

1.2 What Is Borrowed from Where

Feature	Borrowed from	How it works here
Two semesters a year	MIT, Harvard, Princeton; Dhaka	Two 22-week semesters a year

Tutorials	Oxford	Fortnightly oral defence of the student's own problem-set solutions in each course
Problem sets, midterm and final	MIT, Harvard, Princeton	The weekly rhythm of every theory course
Calculus-based core sequence	MIT 8.01–8.04; Harvard 15a/b/c	Years 1 and 2
Collections	Oxford	An ungraded diagnostic test in week 1 on the previous semester
Annual examinations	Oxford Prelims, Parts A, B and C	A comprehensive examination at the end of each year
Viva voce	University of Dhaka (PV 102–402)	A 30–45 minute oral examination each year
Junior Lab and experiment project	MIT 8.13/8.14; Harvard 191; Caltech Ph 77	LAB 301 and LAB 401
Junior Papers	Princeton	IND 301 and IND 302
Communication requirement	MIT CI-M; Caltech Ph 70; Harvard 95	Built into IND 302 and the thesis
Senior Thesis	MIT 8.THU; Princeton; Oxford MPhys	THS 401 and THS 402
Two major options	Oxford Part C (C1–C7)	Major Options I and II in Year 4
Advanced Standing Exams	MIT	Testing out of MTH 101–104 or CMP 101
Honour code	Caltech	Section 2.7

1.3 Document Organisation

The rest of this document consists of the following parts:

- ✎ The rules and regulations of the programme — course system, calendar, grading, examinations, progression and the degree — are presented in Chapter 2.
- ✎ The programme's mission, educational objectives and learning outcomes, and a summary of the whole curriculum, are given in Chapter 3.
- ✎ The details of every course — rationale, learning outcomes, assessment strategy, weekly schedule and texts — are given semester by semester in Chapter 4, with the annual examinations at the end of each year.
- ✎ The Year 4 major options and electives are described in Chapter 5, and the class routines in Chapter 6.

Chapter 2

Rules and Regulations

These are the rules of the department. Where a course syllabus says something different, the syllabus wins for that course. Any change agreed between the student and the instructor is recorded in the changelog of the Programme Handbook in the repository.

2.1 Programme's Course System

Duration and semesters. The programme runs for four academic years. Each year has two semesters of 22 weeks: Spring (February–early July) and Autumn (mid July–mid December). A seven-week, ungraded Orientation Term precedes the first semester.

Load. Four or five courses are taken each semester (alternately 4, 5, 5, 4, 4, 5, 5, 4), at about 4–5 hours each, for a total of 18–22 hours a week. Each course runs over 22 weeks instead of 16, so its total hours are about the same as in the 12-term version. Some weeks before examinations will reach about 25 hours.

Credits. Every course carries 4 credits, and one credit is roughly 25 hours of total work. Each annual examination also carries 4 credits.

Item	Credits	Count	Total
Course (theory, laboratory, computational or independent work)	4	36	144
Annual examination (written papers and viva voce)	4	4	16
		Degree total	**160**

Course types. *Theory* courses (PHY, MTH, options and most electives) have one 80-minute lecture a week and a one-hour recitation every week in a four-course semester, or every other week in a five-course semester. *Computational* courses (CMP) have a lecture and a laboratory. *Laboratory* courses (LAB) replace problem sets with experiments, a notebook and reports. *Independent work* (IND) and the *Senior Thesis* (THS) are supervised research.

Weekly rhythm. Each Monday the week's reading and lectures are released, and each course has a problem set every other week, staggered so that two or three are due each Sunday. Problem sets are due on Sunday at 23:59 (Bangladesh time) and are returned within 48 hours with written feedback, after which the tutorial on that set takes place. Work may be handwritten and uploaded, or committed to the repository under `work/`.

Prerequisites. A course used as a prerequisite must have been passed with C (55%) or better.

Advanced Standing. During the Orientation Term the student may sit a 2-hour Advanced Standing Exam (ASE) in MTH 101, MTH 102, MTH 103, MTH 104 or CMP 101. A score of 70% or more exempts the course; it is recorded as Credit by Examination (CR), carries credits and is not counted in the GPA. The freed slot is filled by an Honours Extra from the electives list.

2.2 Academic Calendar

Every semester follows the same 22-week pattern.

Week	Purpose
1	Teaching starts; *Collection* (ungraded diagnostic) on the previous semester
1–10	Teaching
11	Midterm examinations (teaching continues at a lighter pace)
12–19	Teaching
20	Revision week: no new material; practice papers
21	Final examinations
22	Grading, feedback and the Semester Report; in the second semester of each year, the Annual Examination and viva

Semester	Teaching starts	Midterms	Finals	Ends
Orientation Term	Mon 7 Dec 2026	–	–	Sun 24 Jan 2027
1 Spring 2027	Mon 1 Feb 2027	12 Apr	21 Jun	Sun 4 Jul 2027
2 Autumn 2027	Mon 19 Jul 2027	27 Sep	6 Dec	Sun 19 Dec 2027 (Prelims)
3 Spring 2028	Mon 31 Jan 2028	10 Apr	19 Jun	Sun 2 Jul 2028
4 Autumn 2028	Mon 17 Jul 2028	25 Sep	4 Dec	Sun 17 Dec 2028 (Part A)
5 Spring 2029	Mon 29 Jan 2029	9 Apr	18 Jun	Sun 1 Jul 2029
6 Autumn 2029	Mon 16 Jul 2029	24 Sep	3 Dec	Sun 16 Dec 2029 (Part B)
7 Spring 2030	Mon 4 Feb 2030	15 Apr	24 Jun	Sun 7 Jul 2030
8 Autumn 2030	Mon 22 Jul 2030	30 Sep	9 Dec	Sun 22 Dec 2030 (Finals)

There is a two-week break in July and a winter break of six to seven weeks between the Autumn semester and the next Spring semester. The programme ends on 22 December 2030.

Flexibility. The student's CSE degree comes first when its examinations fall in the same weeks. A semester may be *shifted* by up to two weeks with notice before week 1, or *paused* for up to three weeks mid-semester (for example, for CSE finals or Eid). The calendar is then reissued and the change logged.

2.3 Grading System and Marks Distribution

2.3.1 Letter grades

Letter	Marks (%)	Grade point	Letter	Marks (%)	Grade point
A	85–100	4.0	C+	60–64	2.3
A−	80–84	3.7	C	55–59	2.0
B+	75–79	3.3	D	50–54	1.0
B	70–74	3.0	F	below 50	0.0
B−	65–69	2.7	CR	ASE ≥ 70	not in GPA

Problems are set at MIT, Oxford and Princeton difficulty, so the boundaries are placed where a leading department's curve usually puts them. If an examination turns out harder or easier than intended, the examiner may scale it once, with a written justification recorded in the gradebook.

2.3.2 Marks distribution by course type

Theory courses	
Problem sets	25%
Concept quizzes	5%
Tutorials	10%
Midterm examination	20%
Final examination	40%
Total	100%

Laboratory courses	
Laboratory notebook	15%
Laboratory reports	45%
Oral presentations	15%
Pre-lab quizzes	10%
Final practical	15%
Total	100%

Computational courses (no closed-book exam)	
Coding assignments	35%
Midterm project	20%
Final project	30%
Code review	15%
Total	100%

Junior Papers (IND 301 / IND 302)	
Proposal	10%
Progress meetings / three talks	15 / 25%
Paper	55 / 50%
Presentation / referee report	20 / 15%
Total	100%

The Senior Thesis is weighted as shown on its course pages: in THS 401, the proposal and literature review, progress report, interim presentation and supervisor's assessment carry 25% each; in THS 402, the thesis carries 60%, the oral defence 25% and the poster or public talk 15%. Where an individual course departs from these weights, its own page says so.

2.3.3 How written work is marked

Every problem is marked out of 10.

Criterion	Marks
Physical set-up: principles, diagram, coordinates, stated assumptions	3
Mathematical execution	3
Correct final result, boxed	2
Sanity checks: units, limiting cases, order of magnitude	1
Clarity: someone else could follow the solution	1

A correct answer with no reasoning earns at most 3/10; a clear, correct method with an arithmetic slip earns 8–9/10.

Late work. Late work loses 10% a day for up to three days and then scores zero. Three *late tokens* per semester each give a 24-hour extension with no questions asked. Examinations cannot be late, but they may be rescheduled before the examination week begins.

2.3.4 Tutorials

After each problem set is graded, the tutor picks one or two problems and the student explains the reasoning, in writing or by voice note, and handles follow-up questions (“what changes if…”, “estimate…”). Each tutorial also includes a *self-explanation round*, in which the student talks through the posted solution to the problem found hardest. Each tutorial is scored 0 (absent), 1 (cannot explain own work), 2 (solid) or 3 (insightful).

2.4 Teaching Method

Every lecture, recitation, problem set and tutorial is designed by cognitive load theory (Sweller), as summarised by Scott H. Young. Working memory is small for new material and effectively unlimited for mastered material, and physics has high *element interactivity*, so the aim is to keep the load of each class in the productive zone. The full method is in `docs/teaching-method.md`; in summary:

Principle	What happens in class
Worked-example effect	Every new method is first shown as a fully worked example, then practised: example, completion problem, full problem, mixed problems.
Expertise reversal	Support fades topic by topic; once a method is mastered, the student solves first and sees no example.
Prerequisite fluency	Each lecture opens with a five-minute warm-up on the components it depends on; gaps get a short fluency drill.
Split attention and extraneous load	Integrated notes (labels on figures, equations beside their figures), one primary text, one weekly release.
Goal-free effect	New topics may open with "find everything you can" warm-ups where the possible moves are few.
Germane load	Weekly quizzes (retrieval), self-explanation in tutorials, and interleaved problems once a topic is secure.
Load check	After each lecture and problem set the student rates mental effort from 1 to 9; the rating, with quiz scores, sets the pace.

Problem sets have up to three parts: **A** example pairs (each problem with a solved twin, early in a topic), **B** core problems, and **C** mixed and stretch problems (once the topic is secure). The share of Part A shrinks as each course and the programme go on. The method changes no grading weights.

2.5 Examinations and Progression

Passing a course. A course is passed when the overall weighted mark is at least 50% (grade D) *and* the final-examination mark is at least 40%.

Resit. A student who fails a course, or fails only the final-examination hurdle, may take one resit of the final examination in week 22 or in week 1 of the next semester. The resit replaces the final mark and the course grade is capped at C.

Retake. A course still failed after the resit is retaken in full in a later semester. The new grade replaces the old one in the GPA, but the original F remains visible on the transcript.

Semester standing.

Standing	Rule	Consequence
Dean's List	Semester GPA ≥ 3.70 with no grade below B	Noted on the transcript
Good standing	Semester GPA ≥ 2.00 and every course passed	Continue as planned
Warning	Semester GPA ≥ 2.00 but a course failed	Resit or retake plan agreed in week 22
Probation	Semester GPA < 2.00	Next semester drops to three courses with a remedial plan; two consecutive probations trigger a programme review

Annual examinations. In week 22 of the second semester of each year the student sits an Oxford-style annual examination on the whole year's core courses: *Prelims* (Year 1), *Part A* (Year 2), *Part B* (Year 3) and *Finals, Part C* (Year 4). The written papers carry 80% and the viva voce 20%, and each counts as a 4-credit item in the CGPA. A mark of at least 50% is required to progress to the next year; one resit is allowed in week 1 of the next semester, capped at C. The papers for each year are listed at the end of that year in Chapter 4.

Viva voce. An oral examination, as at Dhaka and Oxford, covering the year's material and the student's own submitted work. The student is asked to derive, explain and estimate. It is marked for correctness (40%), depth of understanding (30%), clarity (15%) and handling of unfamiliar follow-up questions (15%).

Progression. To progress to the next year the student must pass all of that year's courses and its annual examination.

2.6 Degree and Results

GPA. The semester GPA is $\sum(\text{grade points} \times \text{credits}) / \sum \text{credits}$ over the items graded that semester; the CGPA is the same sum over every graded item to date, including annual examinations. The transcript records every course with its code, title, credits, grade, grade points and any resit or retake, together with the SGPA, CGPA, standing and annual-examination results.

Degree requirements. The degree is awarded when the student has (1) passed 160 credits, including all 36 courses or approved substitutes; (2) passed all core courses, with C or better wherever a course was used as a prerequisite; (3) passed all four annual examinations; (4) passed THS 402 Senior Thesis II with C or better; and (5) a CGPA of at least 2.00.

Classification.

CGPA	Classification	US equivalent
≥ 3.85	First Class Honours with Distinction	Summa cum laude
3.70–3.84	First Class Honours	Magna cum laude
3.30–3.69	Upper Second Class (2:1)	Cum laude
2.70–3.29	Lower Second Class (2:2)	–
2.00–2.69	Pass	–

A First requires the Senior Thesis to be graded A− or better; otherwise the classification is capped at 2:1 (Princeton and Harvard practice). If the Year 3–4 CGPA is higher than the overall CGPA, the higher figure is used for classification (Oxford practice).

2.7 Conduct, Submissions and Appeals

Honour code. Caltech's Honor Code reads: *"No member of the Caltech community shall take unfair advantage of any other member of the Caltech community."* Here the only person who can be cheated is the student. Accordingly:

(a) for problem sets, the textbook, lecture notes and the tutor may be used for hints, but not solution manuals, answer websites or another AI's answers to the specific problem before submission (the solved twins in a set's Part A are part of the set and may be used);
(b) examinations are closed book apart from the stated formula sheet, use a simple scientific calculator and no internet, are timed, and have their start and end times written on the first page;
(c) laboratory data must be real; a well-analysed failure scores better than a fabricated success;
(d) papers and the thesis must be the student's own writing, with sources quoted and cited.

A breach scores zero on that item and is noted on the transcript.

Submissions. Work may be uploaded as photographs or a PDF in the course thread, or committed to `work/<semester>/<course>/<item>/` in the repository. Handwritten work is encouraged for problem sets and examinations. Final answers are boxed, with units and limiting cases checked. Code is submitted as `.py` or `.ipynb` files with a short README.

Feedback. Every problem set receives written feedback; there is a mid-semester check-in in week 8; and at the end of each semester the student receives a Semester Report, in the manner of an Oxford college report, with grades, strengths, weaknesses and adjustments for the next semester.

Appeals. A student who believes a mark is wrong says so within seven days. The item is re-marked in full and the decision explained; as at a real university, the re-mark may go up or down.

Chapter 3

OBE and Curriculum Summary

3.1 Programme Mission

To give a computer scientist the education of a physics major from a leading department: a deep, problem-solving command of the core of physics, the mathematics and computation it is written in, the experimental and statistical habits that make it a science, and the experience of original research — learnt on a timetable, assessed honestly, and documented so that it can be shown to others.

3.2 PEO and PLO

Programme Educational Objectives (PEO)

A few years after completing the programme, the graduate is expected to:

- PEO_1 Be prepared for graduate study or research in physics or a neighbouring field, or to bring a physicist's methods to a career in computing.
- PEO_2 Apply physical modelling, computation and quantitative reasoning to complex problems, including those met as a computer scientist.
- PEO_3 Continue to learn independently and act with scientific integrity throughout their careers.

Programme Learning Outcomes (PLO)

On completing the programme, the graduate will be able to:

- PLO_1 Physics Knowledge: Explain and apply the core of physics — classical and quantum mechanics, electromagnetism, thermal and statistical physics, waves, optics and relativity — at the level of a graduating major from a leading department.
- PLO_2 Mathematical Methods: Use calculus, linear algebra, differential equations, complex analysis, Fourier methods, special functions, probability and tensors fluently as tools.
- PLO_3 Problem Solving and Physical Reasoning: Set up and solve unseen problems; check results by units, limiting cases, symmetry and order-of-magnitude estimates.
- PLO_4 Experimental Practice: Design, build and carry out experiments safely, and keep a complete laboratory record.
- PLO_5 Computation: Solve physics problems without closed-form answers using numerical methods, simulation and well-tested code.
- PLO_6 Data Analysis and Uncertainty: Analyse data statistically, fit and test models, and quantify statistical and systematic uncertainty.

PLO_7 Research and Independent Inquiry: Read the research literature, reproduce published results, and carry out an original research project.

PLO_8 Communication: Communicate physics clearly in reports, papers, a thesis, talks and oral examinations.

PLO_9 Modern Physics in Context: Connect the core to modern and frontier physics — relativity, quantum information, particle physics, cosmology, condensed matter — and to other disciplines, especially computer science.

PLO_{10} Independent and Lifelong Learning: Plan and regulate one's own learning, act with integrity, and keep learning beyond the degree.

Mapping of PLOs to PEOs

	PLO_1	PLO_2	PLO_3	PLO_4	PLO_5	PLO_6	PLO_7	PLO_8	PLO_9	PLO_{10}
PEO_1	✓	✓	✓	✓	✓	✓	✓	✓	✓	
PEO_2		✓	✓		✓	✓			✓	
PEO_3							✓	✓		✓

Every course page in Chapter 4 lists its course learning outcomes (CLOs) with the Bloom's taxonomy level and the PLOs each one serves.

3.3 Curriculum at a Glance

Each semester carries four or five 4-credit courses; the second semester of each year also carries the 4-credit annual examination.

Orientation Term — 7 Dec 2026 – 24 Jan 2027 (7 weeks, ungraded)

Course Code	Course Title	Type	Credits	Hours/ Week
ORI 000	Orientation and Bridging	Bridging	0	8–12
		Total	0	

First Semester — Spring 2027 (1 Feb – 4 Jul 2027)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 101	Classical Mechanics I	Theory	4	3
MTH 101	Calculus for Physicists	Theory	4	3
MTH 103	Linear Algebra	Theory	4	3
CMP 101	Computational Physics I: Scientific Python	Computational	4	2.5
		Total	16	

Second Semester

Autumn 2027 (19 Jul – 19 Dec 2027)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 102	Electricity and Magnetism I	Theory	4	3
MTH 102	Multivariable and Vector Calculus	Theory	4	3
MTH 104	Differential Equations	Theory	4	3
LAB 101	Experimental Physics I: Measurement and Uncertainty	Laboratory	4	3.5
PHY 203	Special Relativity	Theory	4	3
EXM 100	Prelims (Annual Examination I)	Exam + viva	4	–
		Total	24	

Third Semester

Spring 2028 (31 Jan – 2 Jul 2028)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 103	Vibrations, Waves and Optics	Theory	4	3
PHY 202	Thermal and Statistical Physics I	Theory	4	3
PHY 204	Classical Mechanics II: Lagrangian and Hamiltonian Dynamics	Theory	4	3
MTH 201	Mathematical Methods: Complex Analysis, PDEs and Special Functions	Theory	4	3
LAB 201	Electronics for Physicists	Laboratory	4	3.5
		Total	20	

Fourth Semester

Autumn 2028 (17 Jul – 17 Dec 2028)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 201	Quantum Physics I	Theory	4	3
PHY 205	Electrodynamics	Theory	4	3
CMP 201	Computational Physics II: Numerical Methods	Computational	4	2.5
PHY 306	Fluids, Nonlinear Dynamics and Complexity	Theory	4	3
EXM 200	Part A (Annual Examination II)	Exam + viva	4	–
		Total	20	

Fifth Semester

Spring 2029 (29 Jan – 1 Jul 2029)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 206	Quantum Physics II	Theory	4	3
LAB 301	Advanced Laboratory I (Junior Lab)	Laboratory	4	4.5
IND 301	Junior Paper I	Independent work	4	4.5
PHY 401	General Relativity and Cosmology	Theory	4	3
		Total	16	

Sixth Semester

Autumn 2029 (16 Jul – 16 Dec 2029)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 301	Quantum Physics III: Approximation Methods and Applications	Theory	4	3
PHY 302	Statistical Mechanics II	Theory	4	3
PHY 402	Quantum Information and Computation	Theory	4	3
IND 302	Junior Paper II and Scientific Communication	Independent work	4	4.5
ELE 4xx	Elective I	Option / elective	4	3
EXM 300	Part B (Annual Examination III)	Exam + viva	4	–
		Total	24	

Seventh Semester

Spring 2030 (4 Feb – 7 Jul 2030)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 303	Atomic, Molecular and Laser Physics	Theory	4	3
PHY 304	Condensed Matter Physics	Theory	4	3
PHY 305	Nuclear and Particle Physics	Theory	4	3
LAB 401	Advanced Laboratory II: Experiment Design Project	Laboratory	4	4.5
THS 401	Senior Thesis I	Research	4	5
		Total	20	

Eighth Semester

Autumn 2030 (22 Jul – 22 Dec 2030)

Course Code	Course Title	Type	Credits	Hours/ Week
OPT 4x1	Major Option I	Option / elective	4	3
OPT 4x2	Major Option II	Option / elective	4	3
ELE 4xx	Elective II	Option / elective	4	3
THS 402	Senior Thesis II and Defence	Research	4	6
EXM 400	Finals (Part C, Annual Examination IV)	Exam + viva	4	–
		Total	20	

3.4 Credit Distribution

The distribution of credits across the categories of course is summarised below.

Category	Courses	Courses	Credits
Core physics (theory)	PHY 101–103, 201–206, 301–306, 401–402	17	68
Mathematics	MTH 101–104, MTH 201	5	20
Computation	CMP 101, CMP 201	2	8
Laboratory	LAB 101, 201, 301, 401	4	16
Independent work and thesis	IND 301, IND 302, THS 401, THS 402	4	16
Specialisation	Two major options and two electives	4	16
Annual examinations	EXM 100, 200, 300, 400	4	16
	Total	**40**	**160**

Chapter 4

Course Details

Detailed outlines of all the courses belonging to the different semesters are produced in the following. Week numbers refer to the 22-week semester of Section 2.2 (they have been converted from the 16-week syllabi of the 12-term version); in theory and computational courses, SECTION A covers the material examined at the midterm and SECTION B the remainder.

4.1 Orientation Term

Course Code	Course Title	Type	Credits	Hours/ Week
ORI 000	Orientation and Bridging	Bridging	0	8–12
		Total	0	

ORI 000

Orientation and Bridging

Ungraded, 8–12 Hours/Week

Prerequisites None

Modelled on MIT and Princeton pre-term placement and bridge programmes; MIT Advanced Standing Exams

Rationale

A good start decides a degree. This ungraded seven-week term measures where the student stands in A-level physics and mathematics, lets prior CSE coursework be recognised through Advanced Standing Exams, closes gaps before the first graded semester, and installs the working habits (problem-solving method, LaTeX, Git, Python, the home lab) that every later course assumes.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Diagnose personal strengths and gaps in school physics and mathematics through timed diagnostics.	Evaluate	PLO_{10}
CLO_2	Apply a systematic problem-solving method: diagram, symbols before numbers, units, limiting cases.	Apply	PLO_3
CLO_3	Use the programme's tools (Git repository, Markdown/LaTeX, Python/Jupyter, phyphox) to submit work.	Apply	PLO_5
CLO_4	Recall and fluently use algebra, trigonometry, calculus and vector toolkits needed for Semester 1.	Remember	PLO_2

Delivery methods and activities include live welcome and bridging lectures, tools laboratories and a mock teaching week.

Assessment tools are timed diagnostics and optional Advanced Standing Exams.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	DIAGNOSTICS D1 AND D2 — ungraded; used for placement and the bridging plan.	–
2	ADVANCED STANDING EXAMS — optional; 70% or more gives credit by examination (CR).	CR
3	MOCK PROBLEM SET AND TUTORIAL — marked with the real rubric for practice; not recorded.	–
	Total	–

Content Outline and Course Schedule

Week	Topic
1	Welcome session; how to study physics; Diagnostic D1 (A-level physics) and D2 (mathematics).
2	Optional Advanced Standing Exams: MTH 101, MTH 102, MTH 103, MTH 104 and CMP 101 (take-home).
3	Placement meeting; tools labs: Git workflow, Markdown/LaTeX, Python and Jupyter, first phyphox measurement.
4	Bridging lectures B1–B2: algebra, trigonometry, exponentials; differentiation and integration toolkit.
5	Bridging lectures B3–B4: vectors; the physicist's method on Morin-style problems. Order books and lab kit.
6	Mock week: mock lectures PHY 101 L0 and MTH 101 L0, mock recitation, quiz, problem set and tutorial.
7	Readiness check; Semester 1 walk-through. Winter break begins.

Texts and Materials

1. G. Pólya. *How to Solve It*. Princeton University Press.
2. D. Morin. *Problems and Solutions in Introductory Mechanics*. CreateSpace.
3. phyphox smartphone laboratory, RWTH Aachen. https://phyphox.org

4.2 First Semester

Spring 2027 (1 Feb – 4 Jul 2027)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 101	Classical Mechanics I	Theory	4	3
MTH 101	Calculus for Physicists	Theory	4	3
MTH 103	Linear Algebra	Theory	4	3
CMP 101	Computational Physics I: Scientific Python	Computational	4	2.5
		Total	16	

PHY 101

Classical Mechanics I

4 Credits, 3 Hours/Week

Prerequisites A-level physics; MTH 101 concurrently

Modelled on MIT 8.01/8.012; Harvard Physics 15a/16; Oxford CP1; Caltech Ph 1; Dhaka PH 101

Rationale

Mechanics is where physics learns to think: choose a system, identify the forces, write the laws, solve, and check. Taught with calculus at the depth of MIT's 8.012, this course builds the habits on which every later course depends, while covering the full Newtonian picture from kinematics to orbits, rigid bodies, rotating frames, oscillations and fluids.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Describe motion in one, two and three dimensions using vectors and Cartesian and polar coordinates.	Understand	PLO_1
CLO_2	Apply Newton's laws with free-body diagrams to systems with constraints, friction and drag.	Apply	PLO_1, PLO_3
CLO_3	Use conservation of momentum, energy and angular momentum to solve collision, orbit and rotation problems.	Apply	PLO_3
CLO_4	Analyse rigid-body motion, non-inertial frames and central-force motion, including Kepler orbits.	Analyze	PLO_1, PLO_3
CLO_5	Model damped and driven oscillators and interpret resonance and the quality factor.	Analyze	PLO_1, PLO_2
CLO_6	Check results by dimensional analysis, limiting cases and order-of-magnitude estimates.	Evaluate	PLO_3

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Vectors, coordinate systems, kinematics in 1D, 2D and 3D; polar coordinates; circular motion.
3	Newton's laws, inertial frames, mass and force; free-body diagrams; constraint equations.
4–5	Friction, drag, tension, pulleys; solving equations of motion; time-varying forces.
6–7	Momentum, centre of mass, systems of particles; variable mass and rockets; impulse.
8	Work and kinetic energy; potential energy, conservative forces, $\mathbf{F} = -\nabla U$.
9–10	Energy diagrams, equilibrium, stability, small oscillations; elastic and inelastic collisions; the CM frame.
11	*Midterm examination* (weeks 1–10).
SECTION B	
12	Angular momentum of a particle, torque; fixed-axis rotation, moment of inertia, parallel-axis theorem.
13–14	Rigid-body dynamics and rolling; gyroscopes and precession.
15	Non-inertial frames: fictitious, centrifugal and Coriolis forces; tides.
16	Central forces, reduced mass, the effective potential; Kepler orbits and Kepler's laws.
17–18	Damped harmonic oscillator; driven oscillator, resonance and Q.
19	Fluids: pressure, buoyancy, continuity, Bernoulli's equation, viscosity.
20–21	Revision; *final examination* (cumulative, 3 hours).

Texts and Materials

1. D. Kleppner and R. Kolenkow. *An Introduction to Mechanics*, 2nd ed. Cambridge University Press. **(Primary text)**
2. D. Morin. *Introduction to Classical Mechanics*. Cambridge University Press.
3. R. P. Feynman, R. B. Leighton and M. Sands. *The Feynman Lectures on Physics*, Vol. I. https://www.feynmanlectures.caltech.edu
4. D. Tong. *Dynamics and Relativity* (lecture notes), University of Cambridge.
5. MIT OpenCourseWare 8.01SC *Classical Mechanics* (video lectures, problem sets, exams).
6. I. E. Irodov. *Problems in General Physics*. (Supplementary problems.)

MTH 101

Calculus for Physicists

4 Credits, 3 Hours/Week

Prerequisites A-level mathematics (Advanced Standing Exam available)

Modelled on MIT 18.01; Harvard Math 1b; Oxford CP3 (part); Caltech Ma 1a; Dhaka Math-102

Rationale

Physics is written in calculus. This course treats single-variable calculus the way a physicist uses it: as a language for rates, accumulation and approximation. It ends with series, complex numbers and first-order differential equations, so that oscillators, waves and quantum mechanics later feel natural rather than foreign.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Compute derivatives and integrals fluently, including by substitution, by parts and by partial fractions.	Apply	PLO_2
CLO_2	Construct linear, quadratic and Taylor approximations and bound their error.	Apply	PLO_2, PLO_3
CLO_3	Determine convergence of sequences and series and manipulate power series.	Analyze	PLO_2
CLO_4	Use complex numbers and Euler's formula to represent oscillations.	Apply	PLO_2
CLO_5	Solve separable and linear first-order ODEs arising in physical models.	Apply	PLO_2, PLO_3

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Limits and continuity; the derivative as a rate of change.
3	Differentiation rules, chain rule; implicit differentiation; inverse, exponential and logarithmic functions.
4–5	Linear and quadratic approximation, Newton's method; curve sketching.
6–7	Optimisation; related rates; the mean value theorem; differentials.
8	Definite integrals, Riemann sums, the fundamental theorem; substitution.
9–10	Applications: areas, volumes, work, centre of mass, averages.
11	*Midterm examination*.
SECTION B	
12	Integration by parts; trigonometric integrals and substitutions.
13–14	Partial fractions; improper integrals; Gaussian and exponential integrals.
15	Sequences and series; convergence tests.
16	Power series and Taylor series; error bounds; physics approximations.
17–18	Complex numbers, polar form, Euler's formula, roots of unity.
19	First-order ODEs: separable and linear; integrating factors; cooling, RC circuits, drag.
20–21	Revision; *final examination*.

Texts and Materials

1. G. Strang. *Calculus*. Wellesley-Cambridge Press (free on MIT OpenCourseWare). **(Primary text)**
2. MIT OpenCourseWare 18.01SC *Single Variable Calculus*.
3. J. Stewart. *Calculus: Early Transcendentals*. Cengage.
4. M. L. Boas. *Mathematical Methods in the Physical Sciences*, 3rd ed. Wiley. (Chapters 1, 2, 8.)
5. T. M. Apostol. *Calculus*, Vol. 1, 2nd ed. Wiley. (Honours reading.)
6. 3Blue1Brown. *Essence of Calculus* (video series).

MTH 103

Linear Algebra

4 Credits, 3 Hours/Week

Prerequisites MTH 101 (Advanced Standing Exam available) *(8-semester version: MTH 101 taken concurrently (Oxford teaches matrices in the first term, CP3).)*

Modelled on MIT 18.06; Harvard Math 21b; Dhaka Math-206

Rationale

Quantum mechanics is linear algebra in disguise. Following Strang's MIT course, the student learns vector spaces, linear maps, orthogonality, eigenvalues and the spectral theorem, ending with complex and function spaces that preview the Hilbert spaces of PHY 201.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Solve linear systems by elimination and interpret the four fundamental subspaces.	Apply	PLO_2
CLO_2	Compute projections and least-squares solutions and relate them to data fitting.	Apply	PLO_2, PLO_6
CLO_3	Find eigenvalues and eigenvectors and diagonalise matrices.	Apply	PLO_2
CLO_4	Explain the spectral theorem for symmetric and Hermitian matrices and its role in physics.	Understand	PLO_2, PLO_1
CLO_5	Represent linear transformations in different bases, including function spaces.	Analyze	PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Vectors, linear combinations, geometry of linear equations.
3	Elimination, LU factorisation, inverses.
4–5	Vector spaces, subspaces, the four fundamental subspaces.
6–7	Independence, basis, dimension, rank.
8	Orthogonality, projections, least squares.
9–10	Gram–Schmidt, QR, orthogonal matrices.
11	*Midterm examination*.
SECTION B	
12	Determinants.
13–14	Eigenvalues and eigenvectors; diagonalisation.
15	Symmetric matrices, spectral theorem, positive definiteness.
16	Complex vectors; Hermitian and unitary matrices; preview of Dirac notation.
17–18	Singular value decomposition; principal components.

Week	Topic
19	Linear transformations, change of basis; function spaces and Fourier bases.
20–21	Revision; *final examination*.

Texts and Materials

1. G. Strang. *Introduction to Linear Algebra*, 6th ed. Wellesley-Cambridge Press. **(Primary text)**
2. MIT OpenCourseWare 18.06SC *Linear Algebra*.
3. S. Axler. *Linear Algebra Done Right*, 4th ed. Springer (open access). (Honours reading.)
4. 3Blue1Brown. *Essence of Linear Algebra* (video series).

CMP 101

Computational Physics I: Scientific Python

4 Credits, 2.5 Hours/Week

Prerequisites None (Advanced Standing Exam available)

Modelled on Harvard Physics 20 / Applied Math 10; MIT 8.006; Dhaka PH 104

Rationale

Most real physics problems have no closed-form solution. Building on the student's programming background, this course turns Python into a physics instrument: numerical calculus, root finding, ODE integrators, Monte Carlo methods, Fourier analysis and data fitting, each applied to a system met in PHY 101 or MTH 101.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Write clear, vectorised NumPy code and produce publication-quality plots.	Apply	PLO_5
CLO_2	Analyse numerical error (round-off, truncation, stability) and choose step sizes accordingly.	Analyze	PLO_5
CLO_3	Implement and compare integrators (Euler, RK4, Verlet) for mechanical systems and check conservation laws.	Evaluate	PLO_5, PLO_3
CLO_4	Fit models to data with χ^2 and report parameter uncertainties.	Apply	PLO_6
CLO_5	Design, validate and present a simulation of a physical system.	Create	PLO_5, PLO_8

Delivery methods and activities include a weekly lecture on the numerical method and a weekly laboratory in which it is coded on a physics problem.

Assessment tools are coding assignments, a midterm project, a final project and a code review.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	CODING ASSIGNMENTS — weekly programs applying the method to a physics problem.	35%
2	MIDTERM PROJECT — weeks 7–8; a short simulation with a written report.	20%
3	FINAL PROJECT — a physics simulation or data analysis with a written report.	30%
4	CODE REVIEW — oral walkthrough of the final project.	15%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Python for scientists: NumPy arrays, vectorisation, plotting trajectories.
3	Floating point, round-off and truncation error; numerical derivatives.
4–5	Numerical integration: trapezoid, Simpson, Gaussian quadrature; period of a pendulum.
6–7	Root finding: bisection, Newton, secant; equilibria and Kepler's equation.
8	ODEs I: Euler's method and its failure; projectile with drag.
9–10	ODEs II: RK4, adaptive steps, `solve_ivp`; damped driven oscillator.
11–12	*Midterm project*: simulate and validate a PHY 101 system.
SECTION B	
13–14	Symplectic integrators (leapfrog, Verlet); planetary orbits.
15	Random numbers and Monte Carlo integration; random walks.
16	Data, uncertainties, least squares and χ^2; fitting g from phyphox data.
17–18	Fourier transforms and the FFT; spectrum of a phone accelerometer signal.
19–20	*Final project*: a simulation or data analysis with a 6–8 page report.
21	Code review and oral walk-through.

Texts and Materials

1. M. Newman. *Computational Physics*. CreateSpace. (Chapters 1–5, 8.) **(Primary text)**
2. *Scientific Python Lecture Notes*. https://scipy-lectures.org
3. NumPy, SciPy and Matplotlib official documentation.

4.3 Second Semester

Autumn 2027 (19 Jul – 19 Dec 2027)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 102	Electricity and Magnetism I	Theory	4	3
MTH 102	Multivariable and Vector Calculus	Theory	4	3
MTH 104	Differential Equations	Theory	4	3
LAB 101	Experimental Physics I: Measurement and Uncertainty	Laboratory	4	3.5
PHY 203	Special Relativity	Theory	4	3
EXM 100	Prelims (Annual Examination I)	Exam + viva	4	–
		Total	24	

PHY 102

Electricity and Magnetism I

4 Credits, 3 Hours/Week

Prerequisites PHY 101, MTH 101; MTH 102 concurrently

Modelled on MIT 8.02/8.022; Harvard Physics 15b; Oxford CP2; Dhaka PH 103

Rationale

Electromagnetism is the first complete field theory a physicist meets and the model for all others. Following Purcell and Morin, the course develops fields, potentials and circuits, shows magnetism emerging from special relativity, and arrives at Maxwell's equations and light, with dielectric and magnetic materials at the end.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Calculate electric fields and potentials of charge distributions using Coulomb's and Gauss's laws.	Apply	PLO_1, PLO_2
CLO_2	Analyse DC, RC and AC circuits, including phasors and resonance.	Analyze	PLO_1
CLO_3	Explain magnetism as a relativistic consequence of moving charges.	Understand	PLO_1
CLO_4	Apply Biot–Savart, Ampère and Faraday laws to compute fields and induced EMFs.	Apply	PLO_1, PLO_3
CLO_5	Derive electromagnetic waves from Maxwell's equations and compute energy flow.	Analyze	PLO_1, PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Charge, Coulomb's law, the electric field, continuous distributions.
3	Flux and Gauss's law; symmetric applications.
4–5	Electric potential; gradient, divergence, Laplacian.
6–7	Conductors, capacitance, field energy; introduction to images.
8	Currents, Ohm's law, DC and RC circuits.
9–10	Fields of moving charges; magnetism as a relativistic effect.
11	*Midterm examination*.
SECTION B	
12	Magnetic field: Biot–Savart, Ampère's law, vector potential.
13–14	Induction: Faraday and Lenz; inductance; magnetic energy.
15	AC circuits, phasors, impedance, resonance.
16	Maxwell's equations, displacement current, EM waves, Poynting vector.
17–18	Dielectrics: polarisation, $\mathbf{D}$, bound charge.
19	Magnetic materials: dia-, para- and ferromagnetism, $\mathbf{H}$, hysteresis.
20–21	Revision; *final examination*.

Texts and Materials

1. E. M. Purcell and D. J. Morin. *Electricity and Magnetism*, 3rd ed. Cambridge University Press. **(Primary text)**
2. MIT OpenCourseWare 8.02 *Physics II: Electricity and Magnetism*; MITx 8.02x.
3. R. P. Feynman et al. *The Feynman Lectures on Physics*, Vol. II.
4. D. Tong. *Electromagnetism* (lecture notes), University of Cambridge.
5. D. Halliday, R. Resnick and J. Walker. *Fundamentals of Physics*. Wiley. (Gentler companion.)

MTH 102

Multivariable and Vector Calculus

4 Credits, 3 Hours/Week

Prerequisites MTH 101 (Advanced Standing Exam available)

Modelled on MIT 18.02; Harvard Math 21a; Oxford CP4 (part); Caltech Ma 1bc; Dhaka Math-103

Rationale

Fields live in three dimensions. This course supplies the calculus of several variables and of vector fields — gradients, multiple integrals in curvilinear coordinates, line and surface integrals and the theorems of Green, Gauss and Stokes — exactly as PHY 102 and later electrodynamics and fluids require.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Compute partial derivatives, gradients and directional derivatives, and optimise with Lagrange multipliers.	Apply	PLO_2
CLO_2	Evaluate multiple integrals in Cartesian, polar, cylindrical and spherical coordinates.	Apply	PLO_2
CLO_3	Evaluate line and surface integrals and identify conservative fields.	Apply	PLO_2
CLO_4	Apply the divergence and Stokes theorems and interpret div and curl physically.	Analyze	PLO_2, PLO_1
CLO_5	Use index notation and ϵ_{ijk} to prove vector identities.	Apply	PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Vectors, dot and cross product, lines and planes, determinants.
3	Matrices and linear systems (introduction); parametric curves, velocity and acceleration.
4–5	Functions of several variables, partial derivatives, tangent planes.
6–7	Chain rule, gradient, directional derivative, multivariable Taylor expansion.
8	Optimisation, second-derivative test, Lagrange multipliers.
9–10	Double integrals; polar coordinates; the Jacobian.
11	*Midterm examination*.
SECTION B	
12	Triple integrals; cylindrical and spherical coordinates; moments of inertia.
13–14	Vector fields, line integrals, work, conservative fields and potentials.
15	Green's theorem; flux in two dimensions.
16	Surface integrals and flux; the divergence theorem.
17–18	Curl and Stokes' theorem; physical meaning of div and curl.
19	Curvilinear grad, div, curl, ∇^2; index notation and ϵ_{ijk}.
20–21	Revision; *final examination*.

Texts and Materials

1. S. J. Colley. *Vector Calculus*. Pearson. **(Primary text)** (Alternative: J. E. Marsden and A. Tromba, *Vector Calculus*, W. H. Freeman.)
2. MIT OpenCourseWare 18.02SC *Multivariable Calculus*.
3. H. M. Schey. *Div, Grad, Curl, and All That*, 4th ed. W. W. Norton.
4. M. L. Boas. *Mathematical Methods in the Physical Sciences*, 3rd ed. Wiley. (Chapters 4–6.)
5. D. Tong. *Vector Calculus* (lecture notes), University of Cambridge.

MTH 104

Differential Equations

4 Credits, 3 Hours/Week

Prerequisites MTH 102; MTH 103 concurrently (Advanced Standing Exam available) *(8-semester version: MTH 102 taken concurrently (as MIT allows for 18.03 with 18.02).)*

Modelled on MIT 18.03; Oxford CP3/CP4; Caltech Ma 2 (part); Dhaka Math-203

Rationale

Laws of physics are differential equations. This course covers linear ODEs and resonance, Fourier series, Laplace transforms, delta functions and Green's functions, linear systems and the matrix exponential, nonlinear phase-plane analysis, and a first treatment of the heat and wave equations.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Solve first- and second-order linear ODEs, including forced and resonant systems.	Apply	PLO_2
CLO_2	Use Fourier series and Laplace transforms to solve driven systems.	Apply	PLO_2
CLO_3	Construct impulse responses and Green's functions using the delta function.	Apply	PLO_2
CLO_4	Analyse linear and nonlinear systems in the phase plane and classify stability.	Analyze	PLO_2, PLO_3
CLO_5	Solve the heat and wave equations by separation of variables.	Apply	PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	First-order ODEs: direction fields, separable and linear equations.
3	Numerical methods; existence and uniqueness.
4–5	Complex exponentials; second-order linear constant-coefficient ODEs.
6–7	Inhomogeneous equations, resonance, frequency response.
8	Fourier series; periodic forcing.
9–10	Delta function, impulse response, convolution, Green's functions.
11	*Midterm examination*.
SECTION B	
12	Laplace transforms.
13–14	Linear systems: eigenvalue methods, the matrix exponential.
15	Phase portraits, stability, linearisation.
16	Nonlinear systems: pendulum, Lotka–Volterra, limit cycles.
17–18	The heat equation by separation of variables.
19	The wave equation; d'Alembert's solution.

Week	Topic
20–21	Revision; *final examination*.

Texts and Materials

1. H. Miller and J. Orloff. MIT 18.03 lecture notes, with M. L. Boas, *Mathematical Methods in the Physical Sciences*, chs. 7–8, 13. **(Primary)**
2. MIT OpenCourseWare 18.03SC *Differential Equations*.
3. M. Tenenbaum and H. Pollard. *Ordinary Differential Equations*. Dover.
4. S. H. Strogatz. *Nonlinear Dynamics and Chaos*, 2nd ed. CRC Press. (Chapters 5–6.)

LAB 101

Experimental Physics I: Measurement and Uncertainty

4 Credits, 3.5 Hours/Week

Prerequisites PHY 101, CMP 101

Modelled on Harvard 15a/b PSI labs; MIT TEAL labs; Oxford first-year practicals; Caltech Ph 3; Dhaka PL 101

Rationale

A number without an uncertainty is not a measurement. Using a smartphone (phyphox), an Arduino and simple apparatus at home, the student learns to measure, estimate and propagate uncertainty, fit models with χ^2, keep a laboratory notebook and write in journal style — the experimental core of every physics degree.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Keep a complete, contemporaneous laboratory notebook.	Apply	PLO_4
CLO_2	Identify, estimate and propagate statistical and systematic uncertainties.	Analyze	PLO_4, PLO_6
CLO_3	Fit physical models to data, test goodness of fit and interpret residuals.	Evaluate	PLO_6
CLO_4	Design and carry out a simple experiment of the student's own.	Create	PLO_4
CLO_5	Report results in journal-style writing and short oral presentations.	Apply	PLO_8

Delivery methods and activities include pre-lab briefings, experiments at home with a smartphone (phyphox), an Arduino and simple apparatus, analysis of open research data, and supervision in office hours.

Assessment tools are the laboratory notebook, reports, oral presentations, pre-lab quizzes and a final practical.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	LABORATORY NOTEBOOK — kept continuously and checked fortnightly.	15%
2	LABORATORY REPORTS — 3–4 reports in journal-article format.	45%
3	ORAL PRESENTATIONS — MIT 8.13-style presentation and questioning on an experiment.	15%
4	PRE-LAB QUIZZES — pre-lab questions and error-analysis exercises.	10%
5	FINAL PRACTICAL — week 15; analysis of unseen data or a build-and-debug practical.	15%
	Total	100%

Content Outline and Course Schedule

Week	Topic
1–2	Safety; the notebook; significant figures; types of error; histograms.
3–5	E1 Pendulum and g: period versus length and amplitude with phyphox; χ^2 fit. Report 1.
6–7	Error propagation, weighted means, Gaussian and Poisson distributions.
8–10	E2 Damped oscillator with phone accelerometer: γ and Q. Report 2.
11	Oral presentation 1.
12–14	E3 RC circuit with Arduino ADC: time constant, calibration, systematics. Report 3.
15	Least squares in depth: residuals, correlated parameters, goodness of fit.
16–18	E4 Magnetic field of a coil or wire with phone magnetometer versus Biot–Savart. Report 4.
19	E5 Mini-experiment of own design (notebook only).
20	Oral presentation 2.
21	*Final practical*: analysis of an unseen dataset (2 hours).

Texts and Materials

1. I. G. Hughes and T. P. A. Hase. *Measurements and their Uncertainties*. Oxford University Press. **(Primary text)**
2. J. R. Taylor. *An Introduction to Error Analysis*, 2nd ed. University Science Books.
3. phyphox experiment documentation. https://phyphox.org

PHY 203

Special Relativity

4 Credits, 3 Hours/Week

Prerequisites PHY 102, MTH 103 *(8-semester version: PHY 102 taken concurrently (special relativity is part of the Oxford Prelims, CP1).)*

Modelled on MIT 8.033/8.20; Oxford CP1 and B2 (part); Harvard 15a (part); Dhaka PH 301 (part)

Rationale

Relativity rebuilds space and time from two postulates. The course develops Lorentz transformations, spacetime diagrams and four-vectors, relativistic dynamics and collisions, the unification of electric and magnetic fields, and — following MIT's 8.033 — a first look at gravity, the Schwarzschild metric and cosmology.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Derive and apply the Lorentz transformation, time dilation and length contraction.	Apply	PLO_1
CLO_2	Resolve relativistic paradoxes using spacetime diagrams and invariants.	Analyze	PLO_1, PLO_3
CLO_3	Use four-vectors to solve collision and decay problems.	Apply	PLO_1, PLO_2
CLO_4	Show how electric and magnetic fields transform between frames.	Understand	PLO_1
CLO_5	Explain the equivalence principle and simple consequences of curved spacetime.	Understand	PLO_1, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Galilean relativity, Michelson–Morley, Einstein's postulates.
3	Simultaneity, time dilation, length contraction, clock synchronisation.
4–5	Lorentz transformation, velocity addition, relativistic Doppler effect.
6–7	Spacetime diagrams, invariant intervals, causality, paradoxes.
8	Four-vectors, proper time, four-velocity and four-momentum.
9–10	Energy and momentum, $E = mc^2$, massless particles.
11	*Midterm examination*.
SECTION B	
12	Collisions, decays, threshold energies, the centre-of-momentum frame.
13–14	Relativistic forces, acceleration, hyperbolic motion, rapidity.
15	Relativity and electromagnetism: transformation of $\mathbf{E}$ and $\mathbf{B}$.
16	Equivalence principle, accelerated frames, gravitational redshift.

Week	Topic
17–18	Schwarzschild metric (introduction): geodesics, light deflection, Shapiro delay, GPS.
19	Newtonian cosmology (introduction).
20–21	Revision; *final examination*.

Texts and Materials

1. D. Morin. *Special Relativity for the Enthusiastic Beginner*. CreateSpace. **(Primary text)**
2. E. F. Taylor and J. A. Wheeler. *Spacetime Physics*, 2nd ed. W. H. Freeman.
3. A. P. French. *Special Relativity*. MIT Introductory Physics Series.
4. A. M. Steane. *Relativity Made Relatively Easy*. Oxford University Press.
5. MIT OpenCourseWare 8.033 *Relativity* and 8.20 *Introduction to Special Relativity*.

EXM 100

Prelims (Annual Examination I)

4 Credits, Papers and Viva

When Week 22 (13–19 Dec 2027)

Modelled on Oxford Prelims (CP1–CP4) and the Dhaka viva voce PV 102.

Paper	Content	Time
Paper 1: Physics	PHY 101, PHY 102, PHY 203 (special relativity is in Year 1 here, as in the Oxford Prelims). Section A: 10 short compulsory questions; Section B: 3 long questions from 6.	3 h
Paper 2: Mathematical Methods	MTH 101–104. Same format.	3 h
Viva voce	The year's physics and the student's own laboratory and computing work.	30 min

Written papers 80%, viva voce 20%. A mark of at least 50% is required to progress to Year 2.

4.4 Third Semester

Spring 2028 (31 Jan – 2 Jul 2028)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 103	Vibrations, Waves and Optics	Theory	4	3
PHY 202	Thermal and Statistical Physics I	Theory	4	3
PHY 204	Classical Mechanics II: Lagrangian and Hamiltonian Dynamics	Theory	4	3
MTH 201	Mathematical Methods: Complex Analysis, PDEs and Special Functions	Theory	4	3
LAB 201	Electronics for Physicists	Laboratory	4	3.5
		Total	20	

PHY 103

Vibrations, Waves and Optics

4 Credits, 3 Hours/Week

Prerequisites PHY 102, MTH 102; MTH 103 and MTH 104 concurrently

Modelled on MIT 8.03; Harvard Physics 15c; Oxford CP4 and A2 (optics); Dhaka PH 201, PH 205

Rationale

Oscillations and waves connect every branch of physics, from strings and sound to light and quantum mechanics. This course develops normal modes, the wave equation, Fourier analysis and dispersion, then applies them to electromagnetic waves, polarisation, interference and diffraction, with two home optics experiments.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Analyse forced, damped and coupled oscillators using complex exponentials and normal modes.	Analyze	PLO_1, PLO_2
CLO_2	Derive and solve the wave equation for continuous systems with boundary conditions.	Apply	PLO_1, PLO_2
CLO_3	Use Fourier series and transforms to describe wave packets, dispersion and group velocity.	Apply	PLO_2
CLO_4	Explain and calculate polarisation, reflection, interference and diffraction phenomena.	Apply	PLO_1
CLO_5	Measure optical quantities at home and compare them with theory.	Evaluate	PLO_4

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Simple harmonic motion, complex exponentials, damping.
3	Forced oscillations, resonance, transients and steady state.
4–5	Coupled oscillators, normal modes, the eigenvalue problem.
6–7	Translation invariance, beaded strings, the continuum limit.
8	The wave equation, standing waves, Fourier series.
9–10	Boundaries, reflection and transmission, impedance matching.
11	*Midterm examination*.
SECTION B	
12	Fourier transforms, wave packets, dispersion, phase and group velocity.
13–14	Waves in 2D and 3D; EM waves from Maxwell's equations; energy and momentum.
15	Polarisation, Fresnel equations, Brewster's angle. Home experiment H1 (Malus's law).
16	Geometric optics, lenses, ray-transfer matrices.
17–18	Interference: two slits, thin films, Michelson interferometer, coherence.
19	Fraunhofer diffraction, gratings, resolution. Home experiment H2 (laser wavelength).
20–21	Revision; *final examination*.

Texts and Materials

1. H. Georgi. *The Physics of Waves*. Prentice Hall (free from the author). **(Primary text)**
2. E. Hecht. *Optics*, 5th ed. Pearson. (Chapters 4–10.) **(Primary text, optics)**
3. A. P. French. *Vibrations and Waves*. MIT Introductory Physics Series.
4. MIT OpenCourseWare 8.03SC *Physics III: Vibrations and Waves*.
5. F. S. Crawford. *Waves* (Berkeley Physics Course, Vol. 3).

PHY 202

Thermal and Statistical Physics I

4 Credits, 3 Hours/Week

Prerequisites PHY 101, MTH 102; PHY 201 concurrently recommended

Modelled on MIT 8.044; Oxford A1; Princeton PHY 301; Caltech Ph 12c; Dhaka PH 102

Rationale

Thermodynamics tells us what is possible; statistical mechanics tells us why. Beginning with probability, the course builds entropy from counting, derives the laws of thermodynamics, treats engines and phase changes, and introduces the Boltzmann factor and quantum statistics.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Use probability distributions and the central limit theorem in physical contexts.	Apply	PLO_2
CLO_2	Apply the laws of thermodynamics to processes, engines and refrigerators.	Apply	PLO_1, PLO_3
CLO_3	Derive entropy, temperature and chemical potential from multiplicity.	Analyze	PLO_1
CLO_4	Use free energies, Maxwell relations and phase-equilibrium conditions.	Apply	PLO_1, PLO_2
CLO_5	Compute partition functions and apply Boltzmann, Fermi–Dirac and Bose–Einstein statistics.	Apply	PLO_1, PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	Problem sets — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	Concept quizzes — a ten-minute quiz at the start of each weekly recitation.	5%
3	Tutorials — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	Midterm examination — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	Final examination — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Probability: random variables, distributions, central limit theorem.
3	Temperature, ideal gases, equipartition; heat, work and the first law.
4–5	Heat capacities, enthalpy; kinetic theory and the Maxwell–Boltzmann distribution.
6–7	The second law: multiplicity, Einstein solids, entropy.
8	Interactions: temperature, pressure and chemical potential from entropy.
9–10	Engines and refrigerators; Carnot, Otto and Diesel cycles.
11	*Midterm examination*.
SECTION B	
12	Free energies, Maxwell relations, thermodynamic identities.
13–14	Phase transformations, Clausius–Clapeyron, van der Waals.
15	The Boltzmann factor, partition function, canonical ensemble.
16	Paramagnetism; equipartition revisited; Maxwell speed distribution.
17–18	Quantum statistics: Fermi–Dirac and Bose–Einstein distributions.
19	Blackbody radiation, degenerate Fermi gases, Debye solids (preview).
20–21	Revision; *final examination*.

Texts and Materials

1. D. V. Schroeder. *An Introduction to Thermal Physics*. Oxford University Press / Addison-Wesley. (Chapters 1–7.) **(Primary text)**
2. S. J. Blundell and K. M. Blundell. *Concepts in Thermal Physics*, 2nd ed. Oxford University Press.
3. C. Kittel and H. Kroemer. *Thermal Physics*, 2nd ed. W. H. Freeman.
4. MIT OpenCourseWare 8.044 *Statistical Physics I*.
5. M. W. Zemansky and R. H. Dittman. *Heat and Thermodynamics*. McGraw-Hill. (Dhaka reference.)

PHY 204

Classical Mechanics II: Lagrangian and Hamiltonian Dynamics

4 Credits, 3 Hours/Week

Prerequisites PHY 101, MTH 104

Modelled on MIT 8.223 and 8.09; Harvard Physics 151; Princeton PHY 205; Caltech Ph 106a; Dhaka PH 301

Rationale

The Lagrangian and Hamiltonian formulations are the bridge from Newton to quantum mechanics and field theory. The course develops variational principles, symmetries and conservation laws, rigid bodies, normal modes, phase space and canonical transformations, and ends with chaos and continuous systems.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Derive equations of motion with the Euler–Lagrange equations, including constraints.	Apply	PLO_1, PLO_2
CLO_2	Relate symmetries to conservation laws through Noether's theorem.	Understand	PLO_1
CLO_3	Analyse rigid-body motion with the inertia tensor and Euler's equations.	Analyze	PLO_1
CLO_4	Formulate Hamiltonian mechanics, Poisson brackets and canonical transformations.	Apply	PLO_1, PLO_2
CLO_5	Investigate nonlinear and chaotic dynamics numerically.	Analyze	PLO_5

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Calculus of variations, Euler–Lagrange equation, brachistochrone, geodesics.
3	Lagrangian mechanics, generalised coordinates, constraints.
4–5	Lagrange multipliers and constraint forces; symmetries and Noether's theorem.
6–7	Two-body problem and central forces; Rutherford scattering.
8	Rigid bodies: inertia tensor, principal axes, Euler's equations, symmetric top.
9–10	Small oscillations: normal-mode matrix method.
11	*Midterm examination*.
SECTION B	
12	Hamiltonian, Hamilton's equations, phase space, Liouville's theorem.
13–14	Poisson brackets, canonical transformations; link to quantum commutators.
15	Hamilton–Jacobi theory, action-angle variables, adiabatic invariants.

Week	Topic
16	Nonlinear dynamics and chaos: driven damped pendulum, Poincaré sections.
17–18	Continuous systems: Lagrangian density for the string.
19	Relativistic Lagrangian; charged particle in an EM field; the Maxwell action.
20–21	Revision; *final examination*.

Texts and Materials

1. J. R. Taylor. *Classical Mechanics*. University Science Books. (Chapters 6–16.) **(Primary text)**
2. D. Tong. *Classical Dynamics* (lecture notes), University of Cambridge.
3. L. N. Hand and J. D. Finch. *Analytical Mechanics*. Cambridge University Press.
4. H. Goldstein, C. Poole and J. Safko. *Classical Mechanics*, 3rd ed. Addison-Wesley. (Reference.)
5. L. D. Landau and E. M. Lifshitz. *Mechanics*, 3rd ed. (Honours reading.)
6. MIT OpenCourseWare 8.223 *Classical Mechanics II* and 8.09 *Classical Mechanics III*.

MTH 201

Mathematical Methods: Complex Analysis, PDEs and Special Functions

4 Credits, 3 Hours/Week

Prerequisites MTH 102, MTH 104

Modelled on MIT 18.04/18.075; Harvard AM 104; Oxford Part A Mathematical Methods; Caltech ACM 95ab; Dhaka PH 203

Rationale

Upper-level physics needs a larger mathematical toolbox: contour integration, Fourier transforms, Sturm–Liouville theory, Legendre and Bessel functions, Green's functions and tensors. This course supplies them one semester before electrodynamics, quantum mechanics and general relativity use them.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Evaluate contour integrals and real integrals using residues.	Apply	PLO_2
CLO_2	Use Fourier transforms to solve diffusion and wave problems.	Apply	PLO_2
CLO_3	Expand functions in orthogonal eigenfunctions from Sturm–Liouville problems.	Apply	PLO_2
CLO_4	Solve Laplace, heat, wave and Schrödinger equations in curvilinear coordinates with special functions.	Apply	PLO_2, PLO_1
CLO_5	Construct Green's functions and manipulate Cartesian tensors.	Analyze	PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Complex functions, analyticity, Cauchy–Riemann equations, harmonic functions.
3	Contour integrals; Cauchy's theorem and integral formula.
4–5	Laurent series, singularities, residues.
6–7	Real integrals by residues; branch cuts.
8	Conformal mapping: 2D electrostatics and potential flow.
9–10	Fourier transforms: convolution, Parseval, diffusion and waves.
11	*Midterm examination*.
SECTION B	
12	Sturm–Liouville theory, orthogonal functions, eigenfunction expansions.
13–14	Legendre polynomials, associated Legendre functions, spherical harmonics.
15	Bessel functions; Laplace's equation in cylindrical and spherical coordinates.
16	Separation of variables for heat, wave and Schrödinger equations in 2D/3D.
17–18	Green's functions for ODEs and PDEs; method of images.
19	Tensors: index notation, summation convention, Cartesian tensors.
20–21	Revision; *final examination*.

Texts and Materials

1. M. L. Boas. *Mathematical Methods in the Physical Sciences*, 3rd ed. Wiley. (Chapters 10–15.) **(Primary text)**
2. K. F. Riley, M. P. Hobson and S. J. Bence. *Mathematical Methods for Physics and Engineering*, 3rd ed. Cambridge University Press.
3. T. Needham. *Visual Complex Analysis*. Oxford University Press.
4. G. B. Arfken, H. J. Weber and F. E. Harris. *Mathematical Methods for Physicists*. Academic Press. (Reference.)
5. MIT OpenCourseWare 18.04 *Complex Variables with Applications*.

LAB 201

Electronics for Physicists

4 Credits, 3.5 Hours/Week

Prerequisites PHY 102, LAB 101

Modelled on Harvard Physics 113/123; Caltech APh/EE 23–24; Dhaka PH 202, PH 403

Rationale

Every modern experiment is read out through electronics. In the spirit of Harvard's laboratory electronics course, the student builds and debugs filters, transistor and op-amp circuits, digital logic, data converters and a feedback controller on a home bench, learning to design instruments rather than merely use them.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Analyse linear circuits in time and frequency domains, including Bode plots.	Analyze	PLO_1, PLO_4
CLO_2	Design and build diode, transistor and op-amp circuits to specification.	Create	PLO_4
CLO_3	Interface sensors with a microcontroller, accounting for sampling and noise.	Apply	PLO_4, PLO_5
CLO_4	Implement a PID feedback controller and characterise its response.	Apply	PLO_4
CLO_5	Debug circuits systematically using a multimeter and oscilloscope.	Evaluate	PLO_4

Delivery methods and activities include pre-lab briefings, experiments at home with a smartphone (phyphox), an Arduino and simple apparatus, analysis of open research data, and supervision in office hours.

Assessment tools are the laboratory notebook, reports, oral presentations, pre-lab quizzes and a final practical.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	LABORATORY NOTEBOOK — kept continuously and checked fortnightly.	15%
2	LABORATORY REPORTS — 3–4 reports in journal-article format.	45%
3	ORAL PRESENTATIONS — MIT 8.13-style presentation and questioning on an experiment.	15%
4	PRE-LAB QUIZZES — pre-lab questions and error-analysis exercises.	10%
5	FINAL PRACTICAL — week 15; analysis of unseen data or a build-and-debug practical.	15%
	Total	100%

Content Outline and Course Schedule

Week	Topic
1–2	DC circuits, dividers, Thévenin and Norton equivalents, instrument loading.
3	RC circuits in the frequency domain: filters, Bode plots, phasors.
4–5	Diodes: rectifiers, clamps, Zener regulation, LEDs.
6–7	BJTs: switch, follower, common-emitter amplifier, biasing.
8	FETs and MOSFET switches; current sources and mirrors.
9–10	Op-amps I: golden rules, inverting and non-inverting amplifiers.
11	*Midterm practical*: design, build and measure against specification.
12	Op-amps II: integrators, differentiators, active filters, non-idealities.
13–14	Oscillators, comparators, Schmitt triggers, 555 timers.
15	Digital logic: gates, flip-flops, counters.
16	ADC/DAC, sampling, aliasing, Arduino interfacing.
17–18	Noise, grounding, shielding; the lock-in amplifier concept.
19	PID feedback control project.
20–21	Project completion; *final practical* (build and debug).

Texts and Materials

1. T. C. Hayes and P. Horowitz. *Learning the Art of Electronics*. Cambridge University Press. **(Primary text)**
2. P. Horowitz and W. Hill. *The Art of Electronics*, 3rd ed. Cambridge University Press.
3. R. L. Boylestad and L. Nashelsky. *Electronic Devices and Circuit Theory*. Pearson. (Dhaka reference.)
4. Falstad circuit simulator; LTspice.

4.5 Fourth Semester

Autumn 2028 (17 Jul – 17 Dec 2028)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 201	Quantum Physics I	Theory	4	3
PHY 205	Electrodynamics	Theory	4	3
CMP 201	Computational Physics II: Numerical Methods	Computational	4	2.5
PHY 306	Fluids, Nonlinear Dynamics and Complexity	Theory	4	3
EXM 200	Part A (Annual Examination II)	Exam + viva	4	–
		Total	20	

PHY 201

Quantum Physics I

4 Credits, 3 Hours/Week

Prerequisites PHY 103, MTH 103, MTH 104

Modelled on MIT 8.04; Harvard Physics 143a; Oxford A3; Princeton PHY 208; Caltech Ph 12b; Dhaka PH 302

Rationale

Quantum mechanics is the course that separates physicists from everyone else. Starting from the experiments that forced it into existence, the student learns wave mechanics, solves the canonical one-dimensional problems and the hydrogen atom, and meets spin — the foundation for all of modern physics.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Explain the experimental evidence for quantisation and wave–particle duality.	Understand	PLO_1
CLO_2	Solve the time-independent Schrödinger equation for standard 1D potentials.	Apply	PLO_1, PLO_2
CLO_3	Use operators, expectation values and the uncertainty principle.	Apply	PLO_1, PLO_2
CLO_4	Solve central-potential problems and the hydrogen atom using separation of variables.	Apply	PLO_1, PLO_2
CLO_5	Describe spin-$\frac{1}{2}$ systems with Pauli matrices.	Understand	PLO_1

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Experimental basis: photoelectric effect, Compton scattering, de Broglie waves, Mach–Zehnder interferometer.
3	Wavefunctions, Born rule, normalisation, expectation values.
4–5	Momentum space, Fourier transforms, the uncertainty principle.
6–7	The Schrödinger equation, stationary states, the infinite well.
8	The harmonic oscillator: analytic and ladder-operator methods.
9–10	Free particle, wave packets, group velocity.
11	*Midterm examination*.
SECTION B	
12	Delta potential, 1D scattering, transmission and reflection, tunnelling.
13–14	Finite well, bound states, introduction to the variational principle.
15	Formalism: Hermitian operators, eigenfunctions, generalised uncertainty principle.
16	3D: separation in spherical coordinates, angular momentum, spherical harmonics.
17–18	The hydrogen atom.
19	Spin-$\tfrac{1}{2}$, Stern–Gerlach, Pauli matrices.
20–21	Revision; *final examination*.

Texts and Materials

1. D. J. Griffiths and D. F. Schroeter. *Introduction to Quantum Mechanics*, 3rd ed. Cambridge University Press. (Chapters 1–4.) **(Primary text)**
2. B. Zwiebach. *Mastering Quantum Mechanics*. MIT Press.
3. A. P. French and E. F. Taylor. *An Introduction to Quantum Physics*. MIT Introductory Physics Series.
4. MIT OpenCourseWare 8.04 *Quantum Physics I* (Zwiebach).
5. R. P. Feynman et al. *The Feynman Lectures on Physics*, Vol. III.

PHY 205

Electrodynamics

4 Credits, 3 Hours/Week

Prerequisites PHY 102, PHY 103, MTH 201

Modelled on MIT 8.07; Harvard Physics 153; Oxford A2; Princeton PHY 304; Caltech Ph 106bc; Dhaka PH 305

Rationale

This is electromagnetism at full mathematical strength: boundary-value problems, multipoles, fields in matter, conservation laws with the stress tensor, waves in media and guides, retarded potentials, radiation, and the covariant formulation that unites it with relativity.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Solve electrostatic boundary-value problems with images, separation of variables and multipoles.	Apply	PLO_1, PLO_2
CLO_2	Analyse fields in dielectric and magnetic media.	Analyze	PLO_1
CLO_3	Apply conservation of energy and momentum to electromagnetic fields.	Apply	PLO_1
CLO_4	Derive wave propagation in vacuum, matter, conductors and waveguides.	Analyze	PLO_1, PLO_2
CLO_5	Calculate radiation from accelerating charges and dipoles; write Maxwell's equations covariantly.	Apply	PLO_1, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	Problem sets — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	Concept quizzes — a ten-minute quiz at the start of each weekly recitation.	5%
3	Tutorials — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	Midterm examination — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	Final examination — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
	SECTION A
1–2	Vector analysis and the Dirac delta; electrostatics revisited formally.
3	Laplace's equation, uniqueness theorems, images.
4–5	Separation of variables in Cartesian, spherical and cylindrical coordinates.
6–7	The multipole expansion.
8	Electric fields in matter: polarisation, $\mathbf{D}$, linear dielectrics.
9–10	Magnetostatics: vector potential, multipoles; magnetic fields in matter.
11	*Midterm examination*.
	SECTION B
12	Electrodynamics: EMF, Faraday, Maxwell's equations in matter.
13–14	Conservation laws: Poynting's theorem, Maxwell stress tensor, field momentum.
15	EM waves in vacuum, matter and conductors; dispersion; waveguides.
16	Potentials and gauges; retarded potentials; Jefimenko; Liénard–Wiechert.
17–18	Radiation: electric and magnetic dipole; Larmor formula.
19	Relativistic electrodynamics: the field tensor and covariant Maxwell equations.
20–21	Revision; *final examination*.

Texts and Materials

1. D. J. Griffiths. *Introduction to Electrodynamics*, 4th/5th ed. Cambridge University Press. **(Primary text)**
2. A. Zangwill. *Modern Electrodynamics*. Cambridge University Press. (Honours reading.)
3. J. D. Jackson. *Classical Electrodynamics*, 3rd ed. Wiley. (Reference.)
4. MIT OpenCourseWare 8.07 *Electromagnetism II*.

CMP 201

Computational Physics II: Numerical Methods

4 Credits, 2.5 Hours/Week

Prerequisites CMP 101, MTH 104

Modelled on Caltech Ph 21/22/121; MIT 8.16 (part); Dhaka PH 307

Rationale

This course gives the student the numerical methods of working physicists: linear algebra and eigenproblems, shooting and relaxation, stable time-stepping of PDEs, spectral methods, Markov-chain Monte Carlo, molecular dynamics and an introduction to machine learning — each applied to a problem from the concurrent courses.

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Solve eigenvalue problems numerically, including bound states of arbitrary potentials.	Apply	PLO_5
CLO_2	Implement stable schemes for elliptic, parabolic and hyperbolic PDEs and analyse stability.	Analyze	PLO_5, PLO_2
CLO_3	Simulate statistical systems with Metropolis Monte Carlo and molecular dynamics.	Apply	PLO_5, PLO_1
CLO_4	Validate numerical results against analytic limits and convergence tests.	Evaluate	PLO_5, PLO_3
CLO_5	Present a computational study in a written report and code review.	Create	PLO_8

Delivery methods and activities include a weekly lecture on the numerical method and a weekly laboratory in which it is coded on a physics problem.

Assessment tools are coding assignments, a midterm project, a final project and a code review.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	CODING ASSIGNMENTS — weekly programs applying the method to a physics problem.	35%
2	MIDTERM PROJECT — weeks 7–8; a short simulation with a written report.	20%
3	FINAL PROJECT — a physics simulation or data analysis with a written report.	30%
4	CODE REVIEW — oral walkthrough of the final project.	15%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Numerical linear algebra: LU, conditioning, sparse systems.
3	Eigenvalue problems: finite-difference Schrödinger solver.
4–5	Shooting method; bound states of arbitrary potentials.
6–7	Elliptic PDEs: Jacobi, Gauss–Seidel and SOR for Laplace and Poisson.
8	Parabolic PDEs: FTCS, stability, Crank–Nicolson; time-dependent Schrödinger.
9–10	Hyperbolic PDEs: the wave equation, the CFL condition.
11–12	*Midterm project*: wave-packet scattering off a barrier.
SECTION B	
13–14	Spectral methods and the FFT for PDEs.
15	Monte Carlo II: importance sampling, Markov chains, Metropolis.

Week	Topic
16	2D Ising model: phase transition, critical temperature, finite-size effects.
17–18	Molecular dynamics: Lennard-Jones gas, thermostats, radial distribution.
19	Optimisation and model fitting; introduction to machine learning for physics data.
20–21	*Final project*; code review and oral.

Texts and Materials

1. M. Newman. *Computational Physics*. CreateSpace. (Chapters 6–10.) **(Primary text)**
2. A. Gezerlis. *Numerical Methods in Physics with Python*, 2nd ed. Cambridge University Press.
3. W. H. Press et al. *Numerical Recipes*, 3rd ed. Cambridge University Press. (Reference.)

PHY 306

Fluids, Nonlinear Dynamics and Complexity

4 Credits, 3 Hours/Week

Prerequisites PHY 204, MTH 201, CMP 201 *(8-semester version: CMP 201 taken concurrently.)*

Modelled on Oxford B1 (Flows, Fluctuations and Complexity); MIT 8.09 and 8.292J; Harvard AM 108

Rationale

Most of the world is nonlinear. The first half of this course is fluid dynamics, from the material derivative to Navier–Stokes, boundary layers and the onset of turbulence; the second half is Strogatz-style nonlinear dynamics — bifurcations, limit cycles, the Lorenz attractor, maps and fractals — ending with random walks, self-organised criticality and networks.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Derive and apply the Euler and Navier–Stokes equations to simple flows.	Apply	PLO_1, PLO_2
CLO_2	Use the Reynolds number and dimensional analysis to classify flow regimes.	Analyze	PLO_3
CLO_3	Classify fixed points and bifurcations in one- and two-dimensional systems.	Analyze	PLO_2
CLO_4	Characterise chaos with Lyapunov exponents, maps and fractal dimensions, analytically and numerically.	Analyze	PLO_5, PLO_2
CLO_5	Explain diffusion, self-organised criticality and network models of complex systems.	Understand	PLO_1, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	The continuum hypothesis, the material derivative, mass conservation.
3	Euler's equations, Bernoulli, hydrostatics.
4–5	Vorticity, Kelvin's circulation theorem, potential flow.
6–7	Viscosity, the Navier–Stokes equations, Poiseuille and Couette flow.
8	The Reynolds number, Stokes flow, boundary layers, drag.
9–10	Water waves, instabilities, turbulence (qualitative; Kolmogorov scaling).
11	*Midterm examination*.
SECTION B	
12	One-dimensional flows and bifurcations: saddle-node, transcritical, pitchfork.
13–14	Two-dimensional flows, the phase plane, limit cycles, the Hopf bifurcation.
15	The Lorenz equations, strange attractors, chaos.
16	One-dimensional maps, the logistic map, period doubling, Feigenbaum's constant.
17–18	Fractals, Lyapunov exponents, chaos in experiments.
19	Complexity: random walks, diffusion, self-organised criticality, introduction to networks.
20–21	Revision; *final examination*.

Texts and Materials

1. D. J. Acheson. *Elementary Fluid Dynamics*. Oxford University Press. (Weeks 1–6.) **(Primary text)**
2. S. H. Strogatz. *Nonlinear Dynamics and Chaos*, 2nd ed. CRC Press. (Weeks 8–13.) **(Primary text)**
3. D. Tong. *Fluid Mechanics* (lecture notes), University of Cambridge.
4. D. J. Tritton. *Physical Fluid Dynamics*. Oxford University Press.

EXM 200

Part A (Annual Examination II)

4 Credits, Papers and Viva

When Week 22 (11–17 Dec 2028)

Modelled on Oxford Part A.

Paper	Content	Time
A1: Waves and Quantum Physics	PHY 103, PHY 201.	3 h
A2: Thermal Physics, Classical Mechanics and Electrodynamics	PHY 202, PHY 204, PHY 205, PHY 306.	3 h
A3: Mathematical and Computational Methods	MTH 201, CMP 201 (written questions only).	2 h
Viva voce	The year's physics and the LAB 201 project.	30 min

Written papers 80%, viva voce 20%. A mark of at least 50% is required to progress to Year 3.

4.6 Fifth Semester

Spring 2029 (29 Jan – 1 Jul 2029)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 206	Quantum Physics II	Theory	4	3
LAB 301	Advanced Laboratory I (Junior Lab)	Laboratory	4	4.5
IND 301	Junior Paper I	Independent work	4	4.5
PHY 401	General Relativity and Cosmology	Theory	4	3
		Total	16	

PHY 206

Quantum Physics II

4 Credits, 3 Hours/Week

Prerequisites PHY 201, MTH 103

Modelled on MIT 8.05; Harvard Physics 143a/b; Oxford A3/B3; Princeton PHY 305; Caltech Ph 125a; Dhaka PH 402

Rationale

The second quantum course recasts the subject in its true language: vector spaces, Dirac notation and operators. With it come two-state systems, entanglement and Bell's theorem, angular-momentum algebra and its addition, and mixed states — the formalism used in all research-level quantum physics and quantum computing.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Use Dirac notation, the spectral theorem and commuting observables.	Apply	PLO_1, PLO_2
CLO_2	Evolve states in the Schrödinger and Heisenberg pictures.	Apply	PLO_1
CLO_3	Analyse two-state dynamics including spin precession and Rabi oscillations.	Analyze	PLO_1
CLO_4	Describe entanglement, test Bell inequalities and explain teleportation.	Understand	PLO_1, PLO_9
CLO_5	Add angular momenta with Clebsch–Gordan coefficients; use density matrices.	Apply	PLO_1, PLO_2

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Two-state systems, Stern–Gerlach, spin as a vector space.
3	Vector spaces, linear operators, inner products, adjoints.
4–5	Dirac notation, spectral theorem, complete sets of commuting observables.
6–7	Uncertainty principle (rigorous) and energy–time relation.
8	Schrödinger and Heisenberg pictures; time evolution.
9–10	Harmonic oscillator algebraically; coherent and squeezed states.
11	*Midterm examination*.
SECTION B	
12	Two-state dynamics: spin precession, NMR, Rabi oscillations.
13–14	Multiparticle states, tensor products, entanglement.
15	EPR, Bell inequalities, teleportation.
16	Angular-momentum algebra; central potentials revisited.
17–18	Addition of angular momentum; Clebsch–Gordan coefficients.
19	Density matrices, mixed states, introduction to decoherence.
20–21	Revision; *final examination*.

Texts and Materials

1. B. Zwiebach. *Mastering Quantum Mechanics*. MIT Press. **(Primary text)**
2. R. Shankar. *Principles of Quantum Mechanics*, 2nd ed. Springer.
3. J. Binney and D. Skinner. *The Physics of Quantum Mechanics*. Oxford University Press.
4. J. S. Townsend. *A Modern Approach to Quantum Mechanics*, 2nd ed. University Science Books.
5. MIT OpenCourseWare 8.05 *Quantum Physics II* (Zwiebach).

LAB 301

Advanced Laboratory I (Junior Lab)

4 Credits, 4.5 Hours/Week

Prerequisites PHY 201, LAB 101, CMP 201

Modelled on MIT 8.13 (Experimental Physics I); Harvard Physics 191; Caltech Ph 77; Princeton PHY 312; Dhaka PL 301

Rationale

Following MIT's Junior Lab, the student performs four experiments in modern physics: two built at home and two that analyse real research data released by CERN and LIGO. Each ends in a journal-style report, and two are defended in 20-minute oral presentations, so that the student learns to measure, analyse and argue like a working experimentalist.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Measure Planck's constant and atomic spectra with home-built apparatus and calibrate against NIST data.	Apply	PLO_4, PLO_6
CLO_2	Reconstruct invariant-mass spectra from CERN open data and fit resonance parameters.	Analyze	PLO_6, PLO_5
CLO_3	Extract a gravitational-wave signal from LIGO data by whitening and matched filtering.	Analyze	PLO_6, PLO_5, PLO_9
CLO_4	Write journal-style reports with complete uncertainty budgets.	Evaluate	PLO_6, PLO_8
CLO_5	Defend experimental work in a 20-minute oral examination.	Evaluate	PLO_8

Delivery methods and activities include pre-lab briefings, experiments at home with a smartphone (phyphox), an Arduino and simple apparatus, analysis of open research data, and supervision in office hours.

Assessment tools are the laboratory notebook, reports, oral presentations, pre-lab quizzes and a final practical.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	LABORATORY NOTEBOOK — kept continuously and checked fortnightly.	15%
2	LABORATORY REPORTS — 3–4 reports in journal-article format.	45%
3	ORAL PRESENTATIONS — MIT 8.13-style presentation and questioning on an experiment.	15%
4	PRE-LAB QUIZZES — pre-lab questions and error-analysis exercises.	10%
5	FINAL PRACTICAL — week 15; analysis of unseen data or a build-and-debug practical.	15%
	Total	100%

Content Outline and Course Schedule

Week	Topic
1–5	J1 Planck's constant from LEDs: turn-on voltage versus wavelength for 4–6 LEDs, wavelengths from the student's own grating spectrometer.
6–10	J2 Atomic spectroscopy: grating spectrometer; mercury and sodium lines; calibration; comparison with NIST; the Rydberg constant.
11	Oral presentation 1.
12–15	J3 Particle physics from CMS open data: dimuon invariant mass; the J/ψ, Υ and Z; fit of the Z mass and width.
16–19	J4 Gravitational waves from LIGO open data (GWOSC): whitening, band-passing, matched filtering of GW150914; chirp mass.
20	Oral presentation 2.
21	*Final practical*: unseen data-analysis practical (3 hours).

Texts and Materials

1. MIT OpenCourseWare 8.13–8.14 *Experimental Physics I & II (Junior Lab)* manuals, and the original papers each cites.
2. P. R. Bevington and D. K. Robinson. *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. McGraw-Hill.
3. CERN Open Data Portal. https://opendata.cern.ch
4. Gravitational Wave Open Science Center (GWOSC). https://gwosc.org
5. NIST Atomic Spectra Database. https://physics.nist.gov/asd

IND 301

Junior Paper I

4 Credits, 4.5 Hours/Week

Prerequisites Year 2 complete

Modelled on Princeton Junior Papers; Harvard Physics 90r; Oxford mini-project

Rationale

Princeton asks every physics junior to write independent papers, and so does this programme. The student chooses an approved topic, reads the primary literature, reproduces a key published result — an analytic derivation or a numerical calculation — and writes it up as a 5,000-word paper in *Physical Review* style, defended in an oral presentation.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Formulate a focused research question and a feasible plan from the primary literature.	Create	PLO_7
CLO_2	Reproduce a published analytic or numerical result and critically assess it.	Evaluate	PLO_7, PLO_3, PLO_5
CLO_3	Write a paper in *Physical Review* style using LaTeX.	Create	PLO_8

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_4	Present and defend the work in a 20-minute talk.	Evaluate	PLO_8

Delivery methods and activities include weekly supervision meetings, written feedback on drafts from the editor, and talks.

Assessment tools are the proposal, progress meetings, the paper and oral presentations.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	Proposal (one page)	10%
2	Weekly progress meetings	15%
3	Paper (about 5,000 words)	55%
4	Oral presentation and questions	20%
	Total	100%

Example topics: the Lorenz attractor and its Lyapunov exponent; Onsager's Ising solution checked numerically; Bohr–Sommerfeld versus exact spectra of anharmonic oscillators; the Kronig–Penney band structure; the Chandrasekhar limit; the Rabi model; the quantum Zeno effect; the fluid dynamics of a draining tank.

Content Outline and Course Schedule

Week	Topic
1–2	Topic discussion; three candidate topics proposed.
3	*Proposal* (one page): question, key references, planned reproduction.
4–8	Literature review; outline due in week 5.
9–15	Reproduction and analysis; weekly 20-minute progress meetings.
16	Complete draft.
17–18	Written feedback from the editor.
19–20	Revision; final paper due at the end of week 14.
21	*Oral presentation* (20 minutes plus questions).

Texts and Materials

1. The primary literature for the chosen topic (*Physical Review*, *American Journal of Physics*, arXiv).
2. American Physical Society. *Physical Review Style and Notation Guide*.
3. N. J. Higham. *Handbook of Writing for the Mathematical Sciences*, 3rd ed. SIAM.

PHY 401

General Relativity and Cosmology

4 Credits, 3 Hours/Week

Prerequisites PHY 203, PHY 204, MTH 201

Modelled on Oxford B5; MIT 8.962, 8.286 and 8.224; Harvard Physics 210; Caltech Ph 236; Dhaka PH 308 (part)

Rationale

Einstein's theory of gravity is taught physics-first, as in Hartle: the student meets the equivalence principle, the Schwarzschild geometry, black holes and gravitational waves before the tensor machinery, then derives the field equations and applies them to the expanding universe, its thermal history and the open problems of dark matter and dark energy.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Compute geodesics and observable effects in the Schwarzschild spacetime.	Apply	PLO_1, PLO_2
CLO_2	Explain black-hole horizons and the generation and detection of gravitational waves.	Understand	PLO_1, PLO_9
CLO_3	Manipulate tensors, covariant derivatives and curvature tensors.	Apply	PLO_2
CLO_4	Derive the Einstein field equations and their Newtonian limit.	Analyze	PLO_1, PLO_2
CLO_5	Solve the Friedmann equations and relate cosmological observations to the thermal history of the universe.	Apply	PLO_1, PLO_6, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	The equivalence principle; gravity as geometry.
3	Metrics, the line element, geodesics from the variational principle.
4–5	The Schwarzschild geometry: orbits, perihelion precession.
6–7	Light bending, gravitational lensing, the Shapiro delay, GPS.
8	Black holes: horizons, Kruskal coordinates, Hawking radiation (qualitative).
9–10	Gravitational waves in linearised gravity; LIGO.
11	*Midterm examination*.
SECTION B	
12	Tensors and differential geometry: covariant derivatives, Christoffel symbols.
13–14	Curvature: the Riemann, Ricci and Einstein tensors.
15	The Einstein field equations; the stress–energy tensor; the Newtonian limit.
16	FRW cosmology, the Friedmann equations, cosmic expansion, the Hubble constant.
17–18	Thermal history: nucleosynthesis, recombination, the cosmic microwave background.
19	Inflation, dark matter, dark energy, open problems.
20–21	Revision; *final examination*.

Texts and Materials

1. J. B. Hartle. *Gravity: An Introduction to Einstein's General Relativity*. Addison-Wesley / Cambridge University Press. (Weeks 1–10.) **(Primary text)**
2. B. Ryden. *Introduction to Cosmology*, 2nd ed. Cambridge University Press. (Weeks 11–13.) **(Primary text)**
3. S. M. Carroll. *Spacetime and Geometry*. Addison-Wesley / Cambridge University Press.
4. B. F. Schutz. *A First Course in General Relativity*, 2nd ed. Cambridge University Press.
5. MIT OpenCourseWare 8.962 *General Relativity* (Hughes) and 8.286 *The Early Universe* (Guth).

4.7 Sixth Semester

Autumn 2029 (16 Jul – 16 Dec 2029)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 301	Quantum Physics III: Approximation Methods and Applications	Theory	4	3
PHY 302	Statistical Mechanics II	Theory	4	3
PHY 402	Quantum Information and Computation	Theory	4	3
IND 302	Junior Paper II and Scientific Communication	Independent work	4	4.5
ELE 4xx	Elective I	Option / elective	4	3
EXM 300	Part B (Annual Examination III)	Exam + viva	4	–
		Total	24	

Elective I is chosen by the end of the Fifth Semester, and the two major options by week 16 of the Sixth Semester; their outlines are in Chapter 5.

PHY 301

Quantum Physics III: Approximation Methods and Applications

4 Credits, 3 Hours/Week

Prerequisites PHY 206

Modelled on MIT 8.06; Harvard Physics 143b; Oxford B3; Princeton PHY 305; Caltech Ph 125b; Dhaka PH 402

Rationale

Almost no real quantum problem can be solved exactly. This course gives the student the approximation methods that make quantum mechanics a working tool — perturbation theory, the variational and WKB methods, time-dependent perturbation theory and scattering — and applies them to real atoms, magnetic fields, radiation and the quantum Hall effect. As in MIT 8.06, a term paper develops scientific writing.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Apply non-degenerate and degenerate perturbation theory to atomic fine and hyperfine structure.	Apply	PLO_1, PLO_2
CLO_2	Estimate ground-state energies with the variational method and tunnelling rates with WKB.	Apply	PLO_2, PLO_3
CLO_3	Derive Fermi's golden rule and use it for absorption, emission and selection rules.	Analyze	PLO_1, PLO_2
CLO_4	Explain Landau levels, the integer quantum Hall effect, the adiabatic theorem and Berry phase.	Understand	PLO_1, PLO_9

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_5	Calculate cross-sections by partial waves and the Born approximation.	Apply	PLO_1, PLO_2
CLO_6	Write a 4,000-word expository paper on an advanced quantum topic.	Create	PLO_8

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

The term paper (4,000 words, MIT 8.06 CI-M model) counts as two problem sets.

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Natural units and the scales of microscopic physics.
3	Non-degenerate perturbation theory.
4–5	Degenerate perturbation theory.
6–7	Hydrogen fine structure: spin–orbit coupling and relativistic corrections.
8	The Zeeman and Stark effects; hyperfine structure.
9–10	The variational method; the helium atom; Born–Oppenheimer approximation.
11	*Midterm examination*.
SECTION B	
12	The WKB approximation, tunnelling, alpha decay.
13–14	Charged particle in a magnetic field: Landau levels, the integer quantum Hall effect.
15	Time-dependent perturbation theory; Fermi's golden rule.
16	Interaction with radiation: absorption, stimulated and spontaneous emission, selection rules.
17–18	The adiabatic theorem and the Berry phase.
19	Scattering: partial waves, phase shifts, the Born approximation, resonances.
20–21	Revision; *final examination*.

Texts and Materials

1. B. Zwiebach. *Mastering Quantum Mechanics*. MIT Press. (Part III.) **(Primary text)**
2. D. J. Griffiths and D. F. Schroeter. *Introduction to Quantum Mechanics*, 3rd ed. Cambridge University Press. (Part II.)
3. J. J. Sakurai and J. Napolitano. *Modern Quantum Mechanics*. Cambridge University Press. (Honours reading.)
4. D. Tong. *Applications of Quantum Mechanics* (lecture notes), University of Cambridge.
5. MIT OpenCourseWare 8.06 *Quantum Physics III* (Zwiebach).

PHY 302

Statistical Mechanics II

4 Credits, 3 Hours/Week

Prerequisites PHY 202, PHY 206

Modelled on MIT 8.08 (honours option 8.333); Harvard Physics 181; Oxford A1/B1; Caltech Ph 127; Dhaka PH 401

Rationale

The second course in statistical physics moves from ideal systems to the collective behaviour of interacting matter. It treats quantum gases in astrophysics and solids, phase transitions and universality, and then the physics of fluctuations and non-equilibrium processes — Brownian motion, master equations and the fluctuation–dissipation theorem — that underlie modern soft-matter and biological physics.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Use the microcanonical, canonical and grand canonical ensembles and compute fluctuations.	Apply	PLO_1, PLO_2
CLO_2	Apply quantum statistics to metals, white dwarfs, phonons, photons and Bose–Einstein condensates.	Apply	PLO_1
CLO_3	Analyse phase transitions with mean-field and Landau theory and explain universality.	Analyze	PLO_1, PLO_9
CLO_4	Derive transport coefficients from kinetic theory and the Boltzmann equation.	Apply	PLO_1, PLO_2
CLO_5	Model stochastic processes with Langevin, Fokker–Planck and master equations.	Apply	PLO_2, PLO_5

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Ensembles revisited: microcanonical, canonical, grand canonical; fluctuations.
3	Ideal quantum gases; the classical limit.
4–5	The degenerate Fermi gas: metals, white dwarfs, the Chandrasekhar limit.
6–7	Bose gases: blackbody radiation, phonons and the Debye model, Bose–Einstein condensation.
8	Interacting gases: the virial expansion; van der Waals revisited.
9–10	Phase transitions: the Ising model, mean-field theory.
11	*Midterm examination*.
SECTION B	
12	Landau theory, critical exponents, universality; the renormalisation group (qualitative).
13–14	Kinetic theory: the Boltzmann equation, mean free path, transport coefficients.
15	Brownian motion, the Langevin equation, Einstein's relation.
16	Fokker–Planck and master equations; Markov processes.
17–18	The fluctuation–dissipation theorem; Johnson–Nyquist noise.
19	Non-equilibrium topics: stochastic thermodynamics and active matter (survey).
20–21	Revision; *final examination*.

Texts and Materials

1. F. Reif. *Fundamentals of Statistical and Thermal Physics*. McGraw-Hill / Waveland Press. **(Primary text)**
2. M. Kardar. *Statistical Physics of Particles*. Cambridge University Press.
3. J. P. Sethna. *Statistical Mechanics: Entropy, Order Parameters, and Complexity*, 2nd ed. Oxford University Press. (Free online.)
4. D. Tong. *Statistical Physics* (lecture notes), University of Cambridge.
5. MIT OpenCourseWare 8.333 *Statistical Mechanics I* (Kardar), selected lectures.

PHY 402

Quantum Information and Computation

4 Credits, 3 Hours/Week

Prerequisites PHY 206

Modelled on MIT 8.370/8.371; Harvard Physics 160; Oxford C2 (part); Caltech Ph 219

Rationale

Quantum information is where physics meets computer science, and where the student's CSE background becomes an advantage. The course develops qubits, circuits and entanglement, the major quantum algorithms and their complexity, noise and error correction, cryptography and physical implementations, and ends with a project run on real IBM Quantum hardware.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Represent qubit states, gates and measurements and build quantum circuits.	Apply	PLO_1, PLO_5
CLO_2	Analyse the Deutsch–Jozsa, Simon, phase-estimation, Shor and Grover algorithms and their complexity.	Analyze	PLO_2, PLO_9
CLO_3	Describe open systems with density matrices and quantum channels.	Apply	PLO_1, PLO_2
CLO_4	Construct and analyse simple quantum error-correcting codes and cryptographic protocols.	Analyze	PLO_1, PLO_3
CLO_5	Implement and evaluate a variational algorithm on real quantum hardware.	Create	PLO_5, PLO_6

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	Problem sets	25%
2	Concept quizzes	5%
3	Tutorials	10%
4	Midterm examination (week 7)	20%
5	Final examination (week 15)	30%
6	Qiskit project on IBM Quantum hardware	10%
	Total	100%

Week	Topic

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Qubits, the Bloch sphere, measurement, multi-qubit states.
3	Gates and circuits; universality.
4–5	Entanglement, Bell/CHSH, teleportation, superdense coding.
6–7	Oracle algorithms: Deutsch–Jozsa, Bernstein–Vazirani, Simon.
8	The quantum Fourier transform and phase estimation.
9–10	Shor's algorithm; the complexity class BQP.
11	*Midterm examination*.
SECTION B	
12	Grover's search; amplitude amplification.
13–14	Density matrices, quantum channels, decoherence.
15	Quantum error correction: bit-flip and Shor codes, stabiliser codes.
16	Quantum cryptography: BB84, no-cloning.
17–18	Physical implementations: superconducting qubits, trapped ions, photons, NV centres.
19	NISQ algorithms: VQE and QAOA; project on IBM Quantum hardware.
20–21	Project write-up; *final examination*.

Texts and Materials

1. M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press. **(Primary text)**
2. J. Preskill. Lecture notes for Caltech Ph 219 *Quantum Computation*.
3. N. D. Mermin. *Quantum Computer Science: An Introduction*. Cambridge University Press.
4. MITx 8.370x *Quantum Information Science*; IBM Qiskit tutorials. https://qiskit.org

IND 302

Junior Paper II and Scientific Communication

4 Credits, 4.5 Hours/Week

Prerequisites IND 301

Modelled on Princeton second Junior Paper; Harvard Physics 95; Caltech Ph 70; MIT CI-M requirement

Rationale

The second Junior Paper is more independent and, ideally, a pilot for the Senior Thesis: it must contain original work, such as a new numerical study, a new analysis of open data or an extension of a published calculation. Alongside it, a communication strand trains the student to give research talks and to referee a published paper.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Carry out a small piece of original research and situate it in the literature.	Create	PLO_7
CLO_2	Explain a recent research paper to a physics audience in a 15-minute talk.	Apply	PLO_8
CLO_3	Write a referee report that evaluates the claims and methods of a published paper.	Evaluate	PLO_8, PLO_7
CLO_4	Write and defend a 6,000-word research paper.	Create	PLO_8, PLO_7

Delivery methods and activities include weekly supervision meetings, written feedback on drafts from the editor, and talks.

Assessment tools are the proposal, progress meetings, the paper and oral presentations.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROPOSAL	10%
2	THREE TALKS	25%
3	PAPER (ABOUT 6,000 WORDS)	50%
4	REFEREE REPORT	15%
	Total	100%

Content Outline and Course Schedule

Week	Topic
1–2	Planning meeting; scope agreed with the supervisor.
3	*Proposal*.
4–5	*Talk 1*: a 15-minute talk explaining a recent arXiv paper.
6–8	Methods and first results.
9–10	*Referee report* on a real published paper.
11	Progress meeting.
12	*Talk 2*: methods and results so far.
13–16	Research; weekly progress meetings.
17–18	Draft paper.
19–20	Revision; final paper due at the end of week 14.
21	*Talk 3*: 20-minute final presentation plus questions.

Texts and Materials

1. The primary literature for the chosen topic (arXiv, *Physical Review*).
2. J.-L. Doumont. *Trees, Maps, and Theorems*. Principiae.
3. American Physical Society. Referee guidelines for *Physical Review* journals.

EXM 300

Part B (Annual Examination III)

4 Credits, Papers and Viva

When Week 22 (10–16 Dec 2029)

Modelled on Oxford Part B.

Paper	Content	Time
B1: Quantum Physics and Quantum Information	PHY 206, PHY 301, PHY 402.	3 h
B2: Statistical Physics, Relativity and Cosmology	PHY 302, PHY 401.	3 h
B3: General Paper	Order-of-magnitude estimates, Fermi problems and short questions across the whole core so far.	2 h
Viva voce	The year's physics and both Junior Papers.	45 min

Written papers 80%, viva voce 20%. A mark of at least 50% is required to progress to Year 4.

4.8 Seventh Semester

Spring 2030 (4 Feb – 7 Jul 2030)

Course Code	Course Title	Type	Credits	Hours/ Week
PHY 303	Atomic, Molecular and Laser Physics	Theory	4	3
PHY 304	Condensed Matter Physics	Theory	4	3
PHY 305	Nuclear and Particle Physics	Theory	4	3
LAB 401	Advanced Laboratory II: Experiment Design Project	Laboratory	4	4.5
THS 401	Senior Thesis I	Research	4	5
		Total	20	

PHY 303

Atomic, Molecular and Laser Physics

4 Credits, 3 Hours/Week

Prerequisites PHY 301, PHY 205

Modelled on Oxford B3; Harvard Physics 175; MIT 8.421 (introductory parts); Dhaka PH 204, PH 306

Rationale

Atoms and light are where quantum mechanics was born and where it is now engineered most precisely. The course builds atomic structure from hydrogen to the periodic table, treats atoms in fields and radiative transitions, explains how lasers work from rate equations to mode locking, and closes with molecules, laser cooling and photonics.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Explain fine, hyperfine and multi-electron structure using LS and jj coupling and Hund's rules.	Understand	PLO_1
CLO_2	Calculate Zeeman, Paschen–Back and Stark shifts.	Apply	PLO_1, PLO_2
CLO_3	Use Einstein coefficients and line-broadening mechanisms to analyse spectra.	Apply	PLO_1, PLO_6
CLO_4	Analyse laser gain, threshold, cavity modes and Gaussian beams.	Analyze	PLO_1, PLO_2
CLO_5	Describe molecular spectra, laser cooling and trapping, and nonlinear optics.	Understand	PLO_1, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Hydrogen revisited: the Bohr model, gross structure, the history of spectroscopy.
3	Fine structure, the Lamb shift (qualitative), hyperfine structure, the 21 cm line.
4–5	Helium, the Pauli principle, exchange.
6–7	Multi-electron atoms: central field, LS and jj coupling, Hund's rules, the periodic table.
8	Atoms in fields: Zeeman (weak and strong field), Paschen–Back, Stark.
9–10	Radiative transitions: Einstein A and B coefficients, selection rules, line broadening.
11	*Midterm examination*.
SECTION B	
12	Lasers I: population inversion, gain, threshold, rate equations.
13–14	Lasers II: optical cavities, Gaussian beams, longitudinal and transverse modes.
15	Laser systems: HeNe, diode, Nd:YAG, Ti:sapphire; pulsed lasers and mode locking.
16	Molecules: Born–Oppenheimer, rotational and vibrational spectra, Raman scattering.
17–18	Laser cooling and trapping, optical tweezers, BEC experiments.
19	Photonics: optical fibres, nonlinear optics (second-harmonic generation), applications.
20–21	Revision; *final examination*.

Texts and Materials

1. C. J. Foot. *Atomic Physics*. Oxford Master Series, Oxford University Press. **(Primary text)**
2. B. H. Bransden and C. J. Joachain. *Physics of Atoms and Molecules*, 2nd ed. Pearson.
3. P. W. Milonni and J. H. Eberly. *Laser Physics*. Wiley.
4. B. E. A. Saleh and M. C. Teich. *Fundamentals of Photonics*. Wiley.

PHY 304

Condensed Matter Physics

4 Credits, 3 Hours/Week

Prerequisites PHY 301, PHY 302

Modelled on Oxford B6; MIT 8.231; Harvard Physics 195; Dhaka PH 303, PH 405

Rationale

Condensed matter is the largest field of physics and the physics behind every semiconductor a computer scientist uses. Following Oxford's approach, the course starts with simple one-dimensional models of phonons and bands before introducing crystal structure, then develops Bloch's theorem, semiconductors and devices, magnetism, and a survey of superconducting and topological matter.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Explain the heat capacity and electrical properties of solids with the Einstein, Debye, Drude and Sommerfeld models.	Understand	PLO_1
CLO_2	Derive phonon and electron band structures for one-dimensional chains and tight-binding models.	Apply	PLO_1, PLO_2
CLO_3	Use reciprocal lattices and Brillouin zones to interpret diffraction experiments.	Apply	PLO_1, PLO_6
CLO_4	Analyse semiconductors, doping and the p–n junction.	Analyze	PLO_1, PLO_4
CLO_5	Describe magnetic order with mean-field theory and survey superconductivity and topological matter.	Understand	PLO_1, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Specific heat of solids: Einstein and Debye models.
3	Electrons in metals: the Drude and Sommerfeld models.
4–5	Chemical bonding; the origin of crystal structure.
6–7	One-dimensional models: phonons in monatomic and diatomic chains; dispersion.
8	One-dimensional tight-binding; the origin of bands.
9–10	Crystal structure: lattices, the unit cell, Miller indices.
11	*Midterm examination*.
SECTION B	
12	The reciprocal lattice, Brillouin zones, X-ray and neutron diffraction.
13–14	The nearly free electron model, band gaps, Bloch's theorem.
15	Insulators, metals and semiconductors; effective mass; holes.
16	Semiconductor devices: doping, the p–n junction, transistors.
17–18	Magnetism: dia- and paramagnetism, ferromagnetism, the Weiss mean-field model, domains.
19	Survey: superconductivity, topological matter, graphene and 2D materials.
20–21	Revision; *final examination*.

Texts and Materials

1. S. H. Simon. *The Oxford Solid State Basics*. Oxford University Press. **(Primary text)**
2. C. Kittel. *Introduction to Solid State Physics*, 8th ed. Wiley.
3. N. W. Ashcroft and N. D. Mermin. *Solid State Physics*. Harcourt. (Reference.)
4. D. Tong. *Solid State Physics* (lecture notes), University of Cambridge.
5. S. H. Simon. Oxford solid state lecture videos; MIT OpenCourseWare 8.231.

PHY 305

Nuclear and Particle Physics

4 Credits, 3 Hours/Week

Prerequisites PHY 301, PHY 203

Modelled on Oxford B4; MIT 8.276/8.701; Harvard Physics 145; Dhaka PH 304, PH 406

Rationale

This course follows matter down to its smallest known constituents. The first half treats the nucleus — binding, the shell model, decays, fission and fusion — and how experiments measure it; the second half builds the Standard Model from relativistic kinematics, symmetries and Feynman diagrams through QCD, the weak interaction and the Higgs mechanism.

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Use the semi-empirical mass formula and the shell model to explain nuclear stability and spins.	Apply	PLO_1
CLO_2	Calculate decay rates for α, β and γ processes and energy release in fission and fusion.	Apply	PLO_1, PLO_2
CLO_3	Compute relativistic kinematics, cross-sections and decay rates.	Apply	PLO_1, PLO_2
CLO_4	Draw and evaluate simple Feynman diagrams and apply conservation laws to particle reactions.	Apply	PLO_1, PLO_3
CLO_5	Explain the structure of the Standard Model, neutrino oscillations and open problems.	Understand	PLO_1, PLO_9

Delivery methods and activities include two 80-minute lectures and one recitation each week, Oxford-style tutorials and office hours; the matching MIT OpenCourseWare lectures serve as a second source.

Assessment tools are problem sets, concept quizzes, tutorials, a midterm and a final examination.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	PROBLEM SETS — about ten weekly sets marked by the ten-point rubric; the lowest is dropped.	25%
2	CONCEPT QUIZZES — a ten-minute quiz at the start of each weekly recitation.	5%
3	TUTORIALS — Oxford-style oral defence of the student's own solutions, scored 0–3.	10%
4	MIDTERM EXAMINATION — week 7; 90 minutes; closed book with a one-sided A4 formula sheet.	20%
5	FINAL EXAMINATION — week 15; 3 hours; cumulative; closed book with a two-sided A4 formula sheet.	40%
	Total	100%

Content Outline and Course Schedule

Week	Topic
SECTION A	
1–2	Nuclear properties: size, mass, binding energy, the semi-empirical mass formula.
3	The nuclear force, the deuteron, isospin.
4–5	The shell model, magic numbers, spin and parity.
6–7	Radioactive decay: α (Gamow tunnelling), β (Fermi theory), γ.
8	Fission, fusion, nucleosynthesis, reactors.
9–10	Detectors and accelerators; how particle physics is measured.
11	*Midterm examination*.

Week	Topic
SECTION B	
12	Relativistic kinematics, cross-sections and decay rates; Fermi's golden rule.
13–14	The particle zoo, the quark model, symmetries and conservation laws.
15	Feynman diagrams; Griffiths' toy theory; QED processes.
16	QCD: colour, confinement, asymptotic freedom, jets.
17–18	Weak interactions, parity and CP violation, neutrino oscillations.
19	Electroweak unification, the Higgs mechanism (qualitative), the Standard Model and beyond.
20–21	Revision; *final examination*.

Texts and Materials

1. D. Griffiths. *Introduction to Elementary Particles*, 2nd ed. Wiley-VCH. **(Primary text)**
2. K. S. Krane. *Introductory Nuclear Physics*. Wiley. (Selected chapters.)
3. B. R. Martin and G. Shaw. *Particle Physics*. Wiley.
4. B. Povh et al. *Particles and Nuclei*. Springer.
5. MIT OpenCourseWare 8.701 *Introduction to Nuclear and Particle Physics*.

LAB 401

Advanced Laboratory II: Experiment Design Project

4 Credits, 4.5 Hours/Week

Prerequisites LAB 301

Modelled on MIT 8.14 (exploratory project); Caltech Ph 77; Harvard Physics 191; Dhaka PH 404, PL 401

Rationale

In the final laboratory course the student stops following manuals. After one guided experiment with a desktop cosmic-ray muon detector, the student designs an experiment of their own: proposing a physics question, budgeting uncertainties, building and debugging the apparatus, taking and analysing data, and defending the result at a poster session, as in MIT 8.14.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Measure the angular distribution of cosmic-ray muons with a scintillator and SiPM detector.	Apply	PLO_4, PLO_6
CLO_2	Propose an experiment with a physics question, expected signal and uncertainty budget.	Create	PLO_4, PLO_7
CLO_3	Build, calibrate and debug original apparatus.	Create	PLO_4
CLO_4	Analyse data critically, including systematic errors, and iterate the design.	Evaluate	PLO_6
CLO_5	Communicate results in a journal-style report, a poster and an oral defence.	Create	PLO_8

Delivery methods and activities include pre-lab briefings, experiments at home with a smartphone (phyphox), an Arduino and simple apparatus, analysis of open research data, and supervision in office hours.

Assessment tools are the laboratory notebook, reports, oral presentations, pre-lab quizzes and a final practical.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	LABORATORY NOTEBOOK	15%
2	GUIDED-EXPERIMENT REPORT	15%
3	PROJECT REPORT	30%
4	ORALS	15%
5	PROPOSAL AND DESIGN REVIEW	10%
6	POSTER AND DEFENCE	15%
	Total	100%

Project ideas: a Michelson interferometer measuring thermal expansion; a Kater reversible pendulum (g to 10^{-4}); a chaotic double pendulum versus simulation; an optical-lever torsion balance; acoustic levitation; a home-made magnetometer tracking geomagnetic variation; Planck's constant from blackbody radiation; laser speckle statistics. The **Senior Thesis proposal** (3–4 pages) is due at the end of week 13 and must be approved before THS 401.

Content Outline and Course Schedule

Week	Topic
1–5	Guided: cosmic-ray muons with a CosmicWatch-style detector; rate versus angle ($\cos^2\theta$) and elevation.
6–7	*Project proposal*: question, apparatus, expected signal, uncertainty budget, risks.
8–10	Design review and build.
11	*Oral*: design review.
12–18	Data taking, analysis, iteration.
19	Poster and report draft. Senior Thesis proposal due.
20	Final report (journal style).
21	*Poster session and oral defence*.

Texts and Materials

1. CosmicWatch desktop muon detector project (MIT). http://www.cosmicwatch.lns.mit.edu
2. P. Horowitz and W. Hill. *The Art of Electronics*, 3rd ed. Cambridge University Press. (Reference.)
3. I. G. Hughes and T. P. A. Hase. *Measurements and their Uncertainties*. Oxford University Press.

THS 401

Senior Thesis I

4 Credits, 5 Hours/Week

Prerequisites IND 302; thesis proposal approved at the end of Semester 10

Modelled on MIT 8.THU; Princeton senior thesis; Oxford MPhys project

Rationale

The senior thesis is the summit of the programme: a year-long piece of original research. It may be computational or theoretical (a numerical study of a model system, or an analytic extension of a published calculation), an open-data analysis (LIGO, CERN, SDSS, TESS/Kepler, Gaia or Planck), or an experiment that extends the LAB 401 project to research quality. The first semester establishes the literature, method and first results.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Review the literature of a research area critically in a written chapter.	Evaluate	PLO_7, PLO_8
CLO_2	Design and carry out a research plan with appropriate methods.	Create	PLO_7, PLO_5, PLO_4
CLO_3	Report progress clearly and respond to supervision.	Apply	PLO_8, PLO_{10}
CLO_4	Present interim results to an examiner.	Evaluate	PLO_8

Delivery methods and activities include weekly supervision meetings, written feedback on chapters and drafts, and presentations.

Assessment tools are written submissions, presentations and an oral defence.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	Proposal and literature review	25%
2	Progress report	25%
3	Interim results presentation	25%
4	Supervisor's assessment of progress (effort, independence, rigour)	25%
	Total	100%

Week	Topic

Content Outline and Course Schedule

Week	Topic
1–2	Kick-off: refine the plan; set up the thesis repository.
3–8	Literature review (written chapter, about 3,000 words).
9–10	*Proposal and literature review* submitted (graded).
11–18	Research; weekly progress meetings with the supervisor.
17–18	*Progress report* (graded).
19–20	Research continues; presentation preparation.
21	*Interim results presentation* (25 minutes plus questions).

Texts and Materials

1. The primary literature for the thesis topic.
2. Open-data portals: GWOSC, CERN Open Data, SDSS, MAST (TESS/Kepler), Gaia Archive, Planck Legacy Archive.
3. U. Alon. “How to choose a good scientific problem”. *Molecular Cell* **35**, 726 (2009).

4.9 Eighth Semester

Autumn 2030 (22 Jul – 22 Dec 2030)

Course Code	Course Title	Type	Credits	Hours/ Week
OPT 4x1	Major Option I	Option / elective	4	3
OPT 4x2	Major Option II	Option / elective	4	3
ELE 4xx	Elective II	Option / elective	4	3
THS 402	Senior Thesis II and Defence	Research	4	6
EXM 400	Finals (Part C, Annual Examination IV)	Exam + viva	4	–
		Total	20	

Elective I is chosen by the end of the Fifth Semester, and the two major options by week 16 of the Sixth Semester; their outlines are in Chapter 5.

THS 402

Senior Thesis II and Defence

4 Credits, 6 Hours/Week

Prerequisites THS 401

Modelled on MIT 8.THU; Princeton senior thesis and departmental exam; Oxford MPhys

Rationale

The second semester completes the research and turns it into a 10,000–15,000-word thesis written in LaTeX, presented publicly and defended orally before the examiner together with the Part C viva. The thesis must earn C or better for the degree to be awarded, and A− or better for First Class Honours.

Course Learning Outcomes and Related Mapping

	Course Learning Outcome (CLO) Upon successful completion of this course, the student will be able to:	Bloom's Level	Associated PLO
CLO_1	Complete an original piece of physics research.	Create	PLO_7
CLO_2	Write a thesis that meets professional standards of rigour, clarity and presentation.	Create	PLO_8
CLO_3	Evaluate the significance and limitations of the work.	Evaluate	PLO_7, PLO_3
CLO_4	Defend the thesis in an oral examination.	Evaluate	PLO_8, PLO_1

Delivery methods and activities include weekly supervision meetings, written feedback on chapters and drafts, and presentations.

Assessment tools are written submissions, presentations and an oral defence.

Assessment strategy is comprised of the following elements, with the distribution of marks shown.

1	Thesis document	60%
2	Oral defence	25%
3	Poster or public talk	15%
	Total	100%

Thesis marking rubric: originality and significance 20% · correctness and rigour 30% · command of the literature 15% · quality of analysis and uncertainty 15% · clarity of writing and figures 20%.

Content Outline and Course Schedule

Week	Topic
1–12	Complete the research.
13–14	Full thesis draft (10,000–15,000 words, LaTeX).
15–16	Editor's feedback; revisions.
17–18	Final checks: figures, references, reproducibility of code and data.
19	*Thesis submitted*.
20	*Poster or public talk*.
22	*Oral defence* (45 minutes), combined with the Part C viva.

Texts and Materials

1. The primary literature for the thesis topic.
2. N. J. Higham. *Handbook of Writing for the Mathematical Sciences*, 3rd ed. SIAM.
3. A. Gopen and J. Swan. "The science of scientific writing". *American Scientist* **78** (1990).

EXM 400

Finals (Part C, Annual Examination IV)

4 Credits, Papers and Viva

When Week 22 (16–22 Dec 2030)

Modelled on Oxford Part C and the Princeton departmental examination.

Paper	Content	Time
C1: General Physics	Synoptic questions across all four years, at the level of a PhD qualifying examination.	3 h
C2: Major Option I	OPT I.	2 h
C3: Major Option II	OPT II.	2 h
C4: Atomic, Condensed Matter, Nuclear and Particle Physics	PHY 303, PHY 304, PHY 305 (taught in Year 4 in this version).	3 h
Viva voce	Combined with the Senior Thesis defence.	45 min

Written papers 80%, viva voce 20%. Once EXM 400 and THS 402 are graded, the degree is classified and the final transcript is issued.

Chapter 5

Major Options and Electives

In Year 4 the student takes two *major options* (Oxford Part C model) and two *electives*, in this version Elective I in the Sixth Semester and the two options and Elective II in the Eighth. The options are chosen by week 16 of the Sixth Semester. Elective I must be one whose prerequisites are met by then: ELE 410, 413, 415, 416, 417 or 419. All options and electives are theory courses with the standard weights (problem sets 25%, quizzes 5%, tutorials 10%, midterm 20%, final 40%) unless stated otherwise, and each option's final examination also forms a Part C paper. A full week-by-week syllabus is issued in the semester before the course runs.

5.1 Major Options

OPT 4A

Astrophysics

4 Credits, 4 Hours/Week

Prerequisites PHY 302, PHY 305; PHY 401 recommended

Modelled on Oxford C1

Texts B. W. Carroll and D. A. Ostlie, *An Introduction to Modern Astrophysics*.

Content

Radiative transfer; stellar structure and the equations of stellar evolution; nuclear burning; white dwarfs, neutron stars and black holes; the interstellar medium; galaxies and galactic dynamics; active galactic nuclei; exoplanet detection (TESS/Kepler data project); large-scale structure.

OPT 4Q

Laser Science, Quantum Optics and Quantum Information Processing

4 Credits, 4 Hours/Week

Prerequisites PHY 303, PHY 402

Modelled on Oxford C2

Texts M. Fox, *Quantum Optics*; C. J. Foot, *Atomic Physics*.

Content

Quantisation of the electromagnetic field; Fock, coherent and squeezed states; the Jaynes–Cummings model and cavity QED; Bloch equations; photon statistics (Hanbury Brown–Twiss); trapped ions and neutral atoms as qubits; photonic quantum computing; quantum metrology.

OPT 4C

Advanced Condensed Matter

4 Credits, 4 Hours/Week

Prerequisites PHY 304

Modelled on Oxford C3

Texts S. M. Girvin and K. Yang, *Modern Condensed Matter Physics*; Ashcroft and Mermin.

Content

Second quantisation; Fermi liquids; BCS superconductivity; magnetism and spin waves; the quantum Hall effects; Berry phase and topological insulators; graphene and 2D materials; a computational project (DFT or tight-binding).

OPT 4P

Particle Physics

4 Credits, 4 Hours/Week

Prerequisites PHY 305

Modelled on Oxford C4

Texts M. Thomson, *Modern Particle Physics*.

Content

The Dirac equation; QED calculations ($e^+e^- \to \mu^+\mu^-$); deep inelastic scattering and partons; QCD; the weak interaction and its V$-$A structure; electroweak unification and the Higgs; neutrino physics; a CMS open-data analysis project.

OPT 4E

Physics of Atmospheres, Oceans and Climate

4 Credits, 4 Hours/Week

Prerequisites PHY 306, PHY 302

Modelled on Oxford C5

Texts G. K. Vallis, *Atmospheric and Oceanic Fluid Dynamics*; R. T. Pierrehumbert, *Principles of Planetary Climate*.

Content

Atmospheric thermodynamics; radiative balance and the greenhouse effect; rotating fluids, geostrophy and Rossby waves; ocean circulation; climate feedbacks and sensitivity; a reanalysis-data project (ERA5).

OPT 4T

Theoretical Physics

4 Credits, 4 Hours/Week

Prerequisites PHY 301, PHY 204, PHY 205; ELE 410 recommended

Modelled on Oxford C6

Texts T. Lancaster and S. J. Blundell, *Quantum Field Theory for the Gifted Amateur*; D. Tong, QFT lecture notes.

Content

Classical field theory and Noether's theorem; canonical quantisation of the scalar field; path integrals in quantum mechanics and field theory; Feynman rules and perturbation theory; the Dirac field; symmetry breaking and the Goldstone theorem; introduction to renormalisation; the link to statistical field theory.

OPT 4B

Biological Physics

4 Credits, 4 Hours/Week

Prerequisites PHY 302, PHY 306

Modelled on Oxford C7

Texts P. Nelson, *Biological Physics*; R. Phillips et al., *Physical Biology of the Cell*.

Content

Life at low Reynolds number; diffusion and random walks in cells; entropic forces and polymer elasticity; molecular motors; membranes and electrostatics in solution; nerve impulses; gene regulation as dynamics; single-molecule data analysis.

5.2 Electives

Code	Elective	Prerequisites	Text	Notes
ELE 410	Mathematical Methods III: group theory, Lie algebras, tensors, differential forms, manifolds	MTH 201, PHY 206	Jeevanjee; Zee	Recommended with OPT 4T
ELE 411	Astrophysics (if not taken as an option)	PHY 302	Ryden and Peterson	–
ELE 412	Plasma Physics	PHY 205, PHY 302	Chen	–
ELE 413	Physics of Energy	PHY 202, PHY 205	Jaffe and Taylor	–
ELE 414	String Theory for Undergraduates	PHY 401, PHY 301	Zwiebach	MIT 8.251 model
ELE 415	Data Science and Machine Learning in Physics	CMP 201	MIT 8.16 modules	Computational weights
ELE 416	Philosophy of Quantum Theory	PHY 301	Albert; Maudlin	3 essays 60%, final paper 40%

Code	Elective	Prerequisites	Text	Notes
ELE 417	History of 20th-Century Physics	–	MIT STS.042J; Pais	Essay-based, as ELE 416
ELE 418	Materials Science and Nanotechnology	PHY 304	Callister; Dhaka PH 407	–
ELE 419	Supervised Reading Course	Instructor’s approval	Agreed reading list	Weekly summaries 50%, essay 50%

Chapter 6

Class Routine

Every term has a fixed weekly class routine, like a university's. A slot is when a class is *scheduled*: the student attends it live with the instructor at that time, or later the same day or week when CSE work clashes. To attend, the student writes the course and lecture number in the course thread (for example, "PHY 101 L5"); the lecture is taught live, and its notes are saved to `lectures/` in the repository. All times are Bangladesh time (UTC+6). The routine for each later semester is issued at least two weeks before it starts, so that it fits around the CSE timetable for that term.

6.1 Default Weekly Pattern

Each theory course has one 80-minute lecture a week and a one-hour recitation (which opens with the quiz) every week in a four-course semester, or every other week in a five-course semester; each computational course has a lecture and a laboratory. Tutorials and office hours are on Saturday afternoon, problem-set blocks on Friday and Saturday mornings, and two or three problem sets (each course every other week) are due on Sunday at 23:59.

6.2 Orientation Term (7 December 2026 – 24 January 2027)

Week	Date and time	Session
1	**Mon 7 Dec, 19:30–20:50**	**O1 Welcome** *(live)*: how the programme works; a Handbook walk-through
	Wed 9 Dec, 19:30–20:50	**O2 How to study physics** *(live)*: Pólya's method, how to read a physics textbook, how to work a problem set
	Fri 11 Dec, 09:00–11:00	**Diagnostic D1: A-level physics** (2 h, timed, closed book)
	Sat 12 Dec, 10:00–12:00	**Diagnostic D2: Mathematics** (2 h, timed, closed book)
2	Mon 14 Dec, 19:30–21:30	*Optional* **ASE MTH 102** (multivariable/vector calculus)
	Mon 14 Dec	*Optional* **ASE CMP 101** take-home coding task released, due Sun 20 Dec 23:59
	Wed 16 Dec, 19:30–21:30	*Optional* **ASE MTH 104** (differential equations)
	Fri 18 Dec, 09:00–11:00	*Optional* **ASE MTH 101** (single-variable calculus and series)
	Sat 19 Dec, 10:00–12:00	*Optional* **ASE MTH 103** (linear algebra)
3	Mon 21 Dec, 19:30–20:30	**Placement meeting** *(live)*: diagnostic and ASE results, exemptions, your bridging plan
	Wed 23 Dec, 19:30–20:50	**Tools lab I** *(live)*: Git and the repo workflow, writing maths in Markdown/LaTeX

Week	Date and time	Session
	Thu 24 Dec, 19:30–20:50	**Tools lab II** *(live)*: Python and Jupyter set-up; your first phyphox measurement
	Sat 26 Dec, 10:00–12:00	Bridging work block (self-study)
4	Mon 28 Dec, 19:30–20:50	**B1 Bridging lecture** *(live)*: algebra, trigonometry and exponential toolkit (adapted to your diagnostic)
	Wed 30 Dec, 19:30–20:50	**B2 Bridging lecture** *(live)*: differentiation and integration toolkit
	Thu 31 Dec, 21:00–22:00	**Bridging recitation** *(live)*
	Sat 2 Jan, 10:00–12:00 · 16:00–16:30	Work block · **first tutorial** (on the bridging exercises)
5	Mon 4 Jan, 19:30–20:50	**B3 Bridging lecture** *(live)*: vectors, dot and cross products, coordinates
	Wed 6 Jan, 19:30–20:50	**B4 Bridging lecture** *(live)*: the physicist's method: diagrams, symbols before numbers, units, limiting cases (Morin, *Problems and Solutions*, ch. 1)
	Thu 7 Jan, 21:00–22:00	**Recitation** *(live)*: Morin-style problems
	Sat 9 Jan, 16:00–16:30	**Tutorial**
	During the week	Order the Year 1 books and the home-lab kit (see the Library)
6	**Mon 11 Jan, 19:30–20:50**	**Mock lecture: PHY 101 "L0"** *(live)*: units, dimensional analysis, estimation
	Wed 13 Jan, 19:30–20:50	**Mock lecture: MTH 101 "L0"** *(live)*: functions and the idea of a limit
	Thu 14 Jan, 21:00–22:00	**Mock recitation and quiz** *(live)*
	Fri 15 Jan, 23:59	**Mock problem set due**, graded with the real rubric
	Sat 16 Jan, 16:00–16:30	**Mock tutorial** on the mock problem set
7	Mon 18 Jan, 19:30–20:30	**Readiness check** (1 h, ungraded)
	Wed 20 Jan, 19:30–20:30	**Semester 1 walk-through and Q&A** *(live)*
	25 – 31 Jan	**Break week.** Semester 1 starts **Mon 1 Feb 2027**

6.3 First Semester, Spring 2027

Four courses over 22 weeks. PHY 101 and MTH 103 problem sets are due on the Sundays of even teaching weeks, MTH 101 sets and CMP 101 assignments on the odd ones.

Weekly routine (teaching weeks 1–10 and 12–19)

Day	Morning	Afternoon	19:30–20:50	21:00–22:20
Mon			PHY 101 Lecture	MTH 101 Lecture
Tue			MTH 103 Lecture	CMP 101 Lecture
Wed			PHY 101 Recitation (19:30–20:30, opens with the quiz)	MTH 101 Recitation (21:00–22:00, opens with the quiz)
Thu			CMP 101 Lab	MTH 103 Recitation (21:00–22:00, opens with the quiz)
Fri	Problem-set block (09:00–12:00)			
Sat	Problem-set block (10:00–12:30)	Tutorials for the two courses whose sets came back (16:00–17:00); office hours (17:30–18:30)		
Sun			Problem-set wrap-up (19:30–21:30)	**Two problem sets due 23:59**

Weekly totals: lectures, recitations and laboratory about 9.7 h; tutorials and office hours 2 h; problem-set blocks 7.5 h; about 19–20 h in all.

Lecture calendar: PHY 101, MTH 101 and MTH 103

Week	Mon	PHY 101, Mon 19:30	MTH 101, Mon 21:00	MTH 103, Tue 19:30
1	Mon 01 Feb	**L1**: Vectors, coordinate systems, kinematics in 1D (K&K ch. 1)	**L1**: Limits and continuity (18.01SC Unit 1; Strang ch. 2)	**L1**: Vectors, linear combinations, the geometry of linear equations (18.06SC Unit I)
2	Mon 08 Feb	**L2**: Kinematics in 2D/3D, polar coordinates, circular motion (K&K ch. 1)	**L2**: The derivative as a rate of change (18.01SC Unit 1; Strang ch. 2)	**L2**: Elimination and matrix multiplication (18.06SC Unit I)
3	Mon 15 Feb	**L3**: Newton's laws, inertial frames, free-body diagrams (K&K ch. 2)	**L3**: Differentiation rules, the chain rule (18.01SC Unit 1; Strang ch. 4)	**L3**: LU factorisation, inverses, transposes and permutations (18.06SC Unit I)
4	Mon 22 Feb	**L4**: Friction, drag, tension, constraints; solving equations of motion (K&K ch. 3)	**L4**: Implicit differentiation; inverse, exponential and log functions (18.01SC Unit 1; Strang ch. 4, 6)	**L4**: Vector spaces and subspaces; column space and null space (18.06SC Unit I)
5	Mon 01 Mar	**L5**: Momentum, centre of mass, systems of particles (K&K ch. 4)	**L5**: Linear and quadratic approximation, Newton's method (18.01SC Unit 2; Strang ch. 3)	**L5**: Solving $Ax = 0$ and $Ax = b$; the four fundamental subspaces (18.06SC Unit I)
6	Mon 08 Mar	**L6**: Variable mass, rockets, impulse (K&K ch. 4)	**L6**: Curve sketching and optimisation (18.01SC Unit 2; Strang ch. 3)	**L6**: Independence, basis, dimension, rank (18.06SC Unit I)
7	Mon 15 Mar	**L7**: Work, kinetic energy, the work–energy theorem (K&K ch. 5)	**L7**: Related rates, the mean value theorem, differentials (18.01SC Unit 2; Strang ch. 3)	**L7**: Orthogonality and projections (18.06SC Unit II)
8	Mon 22 Mar	**L8**: Potential energy, conservative forces, $F = -\nabla U$ (K&K ch. 5)	**L8**: Definite integrals, Riemann sums, the fundamental theorem (18.01SC Unit 3; Strang ch. 5)	**L8**: Least squares (links to CMP 101 fitting and LAB 101) (18.06SC Unit II)

Week	Mon	PHY 101, Mon 19:30	MTH 101, Mon 21:00	MTH 103, Tue 19:30
9	Mon 29 Mar	**L9**: Energy diagrams, equilibrium, stability, small oscillations (K&K ch. 6)	**L9**: Substitution (18.01SC Unit 3; Strang ch. 5)	**L9**: Gram–Schmidt, QR, orthogonal matrices (18.06SC Unit II)
10	Mon 05 Apr	**L10**: Collisions: elastic, inelastic, the CM frame (K&K ch. 6)	**L10**: Areas, volumes, work, centre of mass, averages (18.01SC Unit 3; Strang ch. 8)	**L10**: Determinants (18.06SC Unit II)
11	Mon 12 Apr	**Midterm week** (see below)		
12	Mon 19 Apr	**L11**: Angular momentum, torque, fixed-axis rotation, moment of inertia (K&K ch. 7)	**L11**: Integration by parts; trigonometric integrals and substitutions (18.01SC Unit 4; Strang ch. 7)	**L11**: Eigenvalues and eigenvectors (18.06SC Unit II)
13	Mon 26 Apr	**L12**: Rigid-body dynamics, rolling motion (K&K ch. 7, 8)	**L12**: Partial fractions; improper integrals, Gaussian integrals (18.01SC Unit 4; Strang ch. 7)	**L12**: Diagonalisation, powers of matrices, systems of linear ODEs (18.06SC Unit II)
14	Mon 03 May	**L13**: Gyroscopes and precession (K&K ch. 8)	**L13**: Sequences, series and convergence tests (18.01SC Unit 5; Strang ch. 10)	**L13**: Symmetric matrices and the spectral theorem (18.06SC Unit III)
15	Mon 10 May	**L14**: Non-inertial frames: centrifugal and Coriolis forces, tides (K&K ch. 9)	**L14**: Power series and Taylor series (18.01SC Unit 5; Strang ch. 10)	**L14**: Positive definite matrices (18.06SC Unit III)
16	Mon 17 May	**L15**: Central forces, reduced mass, the effective potential (K&K ch. 10)	**L15**: Taylor error bounds and physics approximations (18.01SC Unit 5; Boas 1)	**L15**: Complex vectors, Hermitian and unitary matrices, Dirac notation preview (18.06SC Unit III)
17	Mon 24 May	**L16**: Kepler orbits and Kepler's laws (K&K ch. 10)	**L16**: Complex numbers, polar form, Euler's formula (Boas 2.1–2.11)	**L16**: The SVD and principal components (18.06SC Unit III)
18	Mon 31 May	**L17**: Damped and driven oscillators, resonance, Q factor (K&K ch. 11)	**L17**: Complex exponentials for oscillations, roots of unity (Boas 2.12–2.17)	**L17**: Linear transformations and change of basis (18.06SC Unit III)
19	Mon 07 Jun	**L18**: Fluids: pressure, buoyancy, continuity, Bernoulli, viscosity (8.01SC fluids notes; DU PH 101 §8–9)	**L18**: First-order ODEs: separable and linear, integrating factors (Boas 8.1–8.3)	**L18**: Function spaces: Fourier series as an orthogonal basis (18.06SC Unit II)

Lecture calendar: CMP 101

Week	Lecture (Tue) / Lab (Thu)	Method	Physics application
1	Tue 02 Feb / Thu 04 Feb	Python for scientists: NumPy arrays, vectorisation	Plotting trajectories, unit handling
2	Tue 09 Feb / Thu 11 Feb	Plotting, notebooks and Git for code	Visualising PHY 101 kinematics
3	Tue 16 Feb / Thu 18 Feb	Floating point, round-off and truncation error	Numerical derivatives of x(t)
4	Tue 23 Feb / Thu 25 Feb	Numerical integration: trapezoid and Simpson	Work by variable forces
5	Tue 02 Mar / Thu 04 Mar	Gaussian quadrature and error estimates	Period of a pendulum at large amplitude
6	Tue 09 Mar / Thu 11 Mar	Root finding: bisection, Newton, secant	Equilibria; Kepler's equation
7	Tue 16 Mar / Thu 18 Mar	ODEs I: Euler's method and why it fails	Projectile with drag
8	Tue 23 Mar / Thu 25 Mar	ODEs II: RK4, adaptive steps, solve_ivp	Damped driven oscillator
9	Tue 30 Mar / Thu 01 Apr	Linear algebra in NumPy: solving systems, eigenvalues	Normal modes of coupled oscillators (with MTH 103)

Week	Lecture (Tue) / Lab (Thu)	Method	Physics application
10	Tue 06 Apr / Thu 08 Apr	Midterm project kickoff	Your PHY 101 simulation
11	–	Midterm week: project work (no CMP sessions)	
12	Tue 20 Apr / Thu 22 Apr	Midterm project clinic	Your PHY 101 simulation
13	Tue 27 Apr / Thu 29 Apr	Symplectic integrators (leapfrog, Verlet)	Planetary orbits
14	Tue 04 May / Thu 06 May	Random numbers, Monte Carlo integration	Estimating π; volumes
15	Tue 11 May / Thu 13 May	Random walks and diffusion	Brownian motion
16	Tue 18 May / Thu 20 May	Data, uncertainties, least squares, χ^2	Fitting g from phyphox data
17	Tue 25 May / Thu 27 May	Fourier transforms and the FFT	Spectrum of your phone's accelerometer
18	Tue 01 Jun / Thu 03 Jun	Final project kickoff	Choose and scope your project
19	Tue 08 Jun / Thu 10 Jun	Final project clinic	Your project

Examination and special weeks

Week	Dates	What happens
1	Mon 1 Feb	Teaching starts. No Collection this semester because it's the first one
11	12–18 Apr	**Midterm week. PHY 101 midterm Mon 12 Apr, 19:30–21:00. MTH 103 midterm Tue 13 Apr, 19:30–21:00. MTH 101 midterm Wed 14 Apr, 19:30–21:00.** CMP 101 midterm project work. No problem sets due
12	19–25 Apr	CMP 101 midterm project due Sun 25 Apr, 23:59
20	14–20 Jun	**Revision week.** Lecture slots become review lectures. Recitations become practice-paper sessions. CMP 101 final project due Sun 20 Jun
21	21–27 Jun	**Finals. PHY 101 final Mon 21 Jun, 18:30–21:30. MTH 103 final Tue 22 Jun, 18:30–21:30. MTH 101 final Wed 23 Jun, 18:30–21:30. CMP 101 code review Thu 24 Jun, 19:30**
22	28 Jun – 4 Jul	Grades, written feedback and your **Semester Report**. Two-week break, then S2 starts on Mon 19 Jul

Ramadan and Eid. Ramadan 2027 is expected to run from about 8 February to 9 March (weeks 2–6), with Eid al-Fitr around 10 March (week 6), subject to the moon sighting; the midterms are five weeks later. If needed, evening slots can move to 21:30–23:30 after tarawih, or a one-week Eid pause can be taken in week 6, absorbed by shortening the July break to one week.

Five-course semesters. In S2, S3, S6 and S7 each theory course keeps its weekly lecture but has a recitation every other week, alternating with another course, so that classes stay at about 10 hours a week.